EE-102: Electric Circuit Analysis

LECTURE NO 9: *AC Fundamentals*

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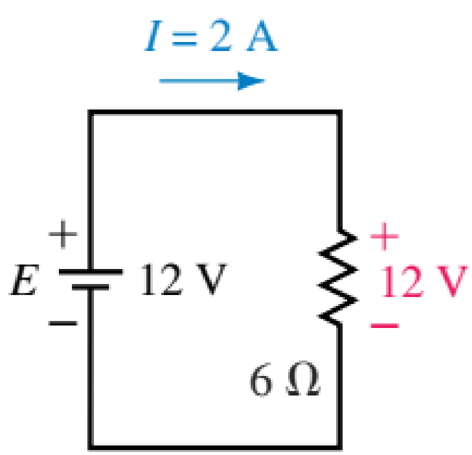
1. Introduction (1/2)

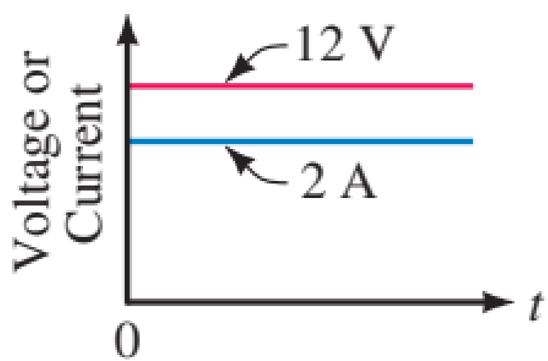
- ***Alternating current (ac)*** is the current that alternates (changes) in direction (usually many times per second).
- It passes first in one direction then in other through the circuit. Such currents are produced by voltage sources whose polarities alternate b/w positive and negative (rather than being fixed as with dc sources).
- Conventionally, alternating currents are called ***ac currents*** and alternating voltages are called ***ac voltages***.
- Variation of an ac voltage or current versus time is called its waveform.
- *v(t), i(t) and e(t)* and so on, are instantaneous quantities.

1. Introduction (2/2)

- DC sources have fixed polarities and constant magnitudes and thus producing currents with constant value and unchanging direction.
- In contrast, the voltages of ac sources alternate in polarity and vary in magnitude, thus producing currents that vary in magnitude and alternate in direction.



(a)

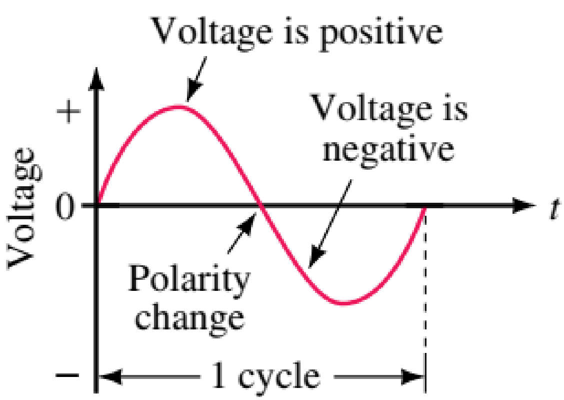


(b) Voltage and current versus time for dc

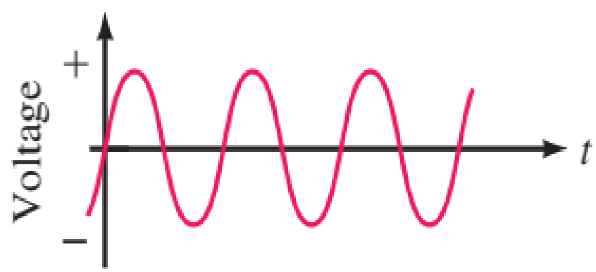
In a dc circuit, voltage polarities and current directions do not change.

2. Sinusoidal AC voltage

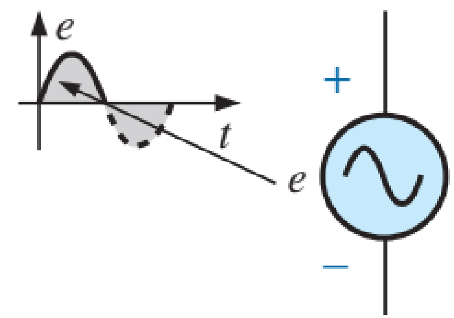
- Voltage at the wall outlet in your home.(***sine wave*** or ***sinusoidal ac waveform***).
- One complete variation is called ***cycle***.
- Since the wave repeats itself at regular intervals as in (b), it is called a ***periodic*** waveform.



(a) Variation of voltage versus time



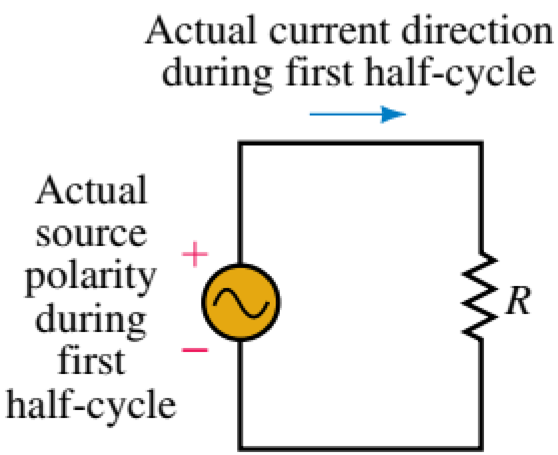
(b) A continuous stream of cycles



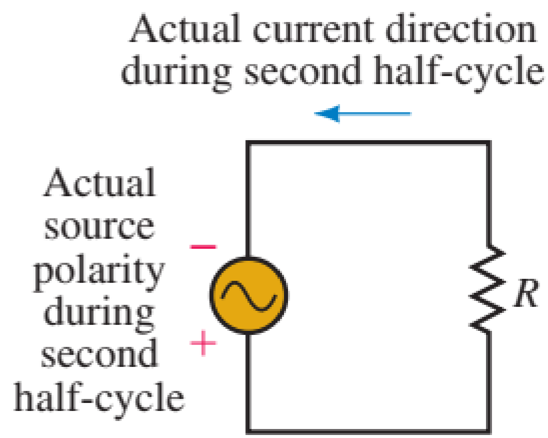
(c) Sinusoidal ac voltage source symbol

3. Sinusoidal AC current

- Since current is proportional to voltage, its shape is also sinusoidal.



(a)



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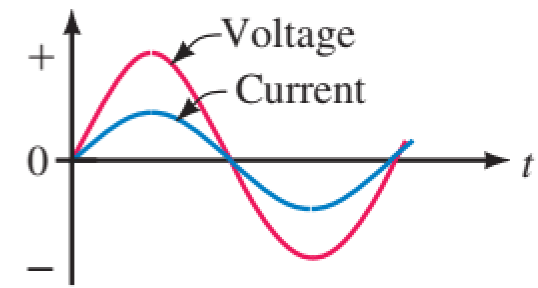
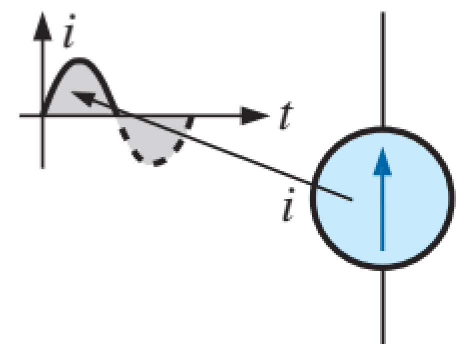


(c)

In (a), First Half-Cycle = Positive Half Cycle.
In (b), Second Half-Cycle = Negative Half Cycle.

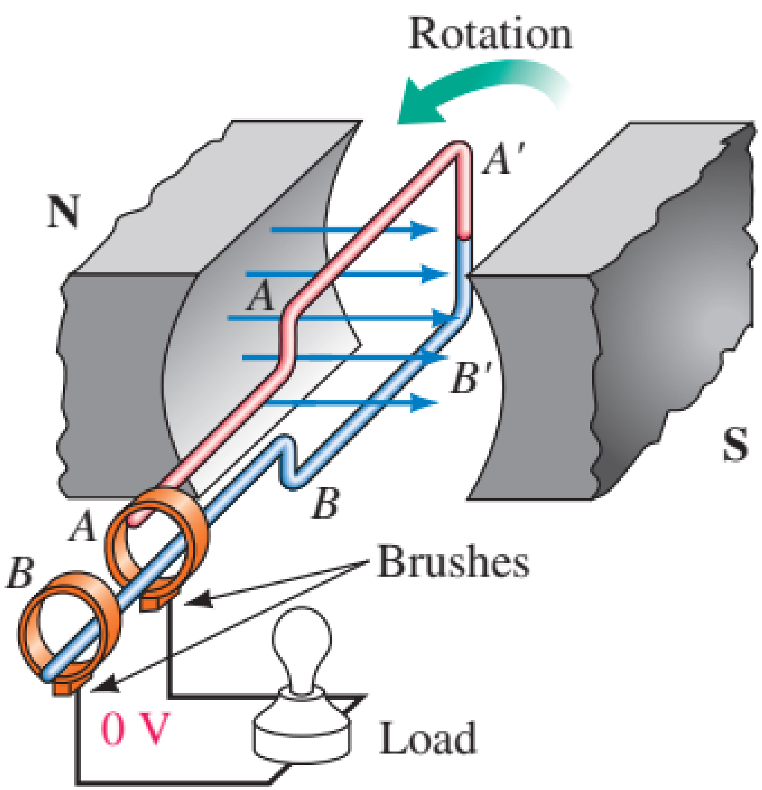


(d) Sinusoidal ac current source symbol

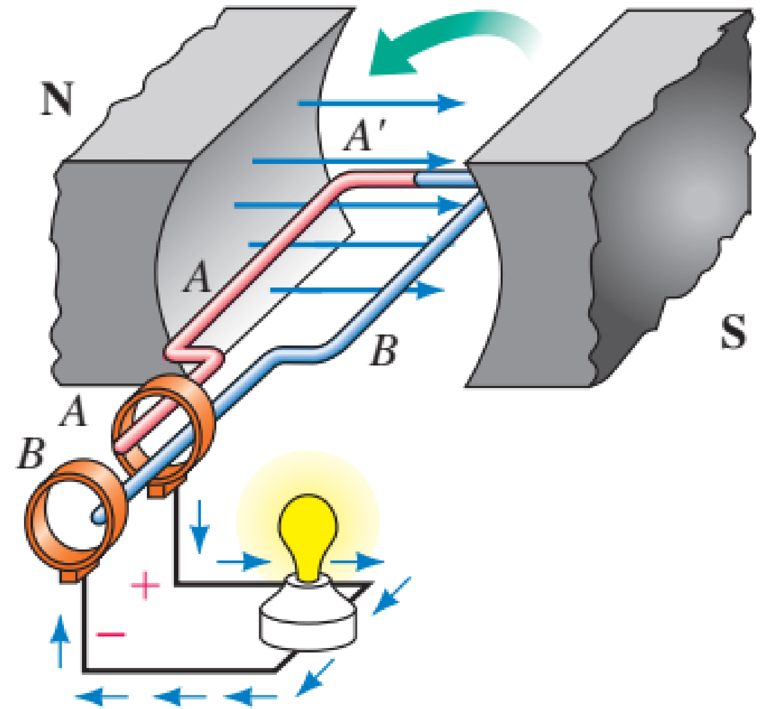
4. Generating ac voltages (1/4)

- One way to generate an ac voltage is to rotate a coil of wire at constant angular velocity in a fixed magnetic field as shown in figure(s) next slide.
- Slip rings and brushes connect the coil to the load.
- Faradays law revisited. "*The rate of change of flux is directly proportional to induced voltage*".

4. Generating ac voltages (2/4)

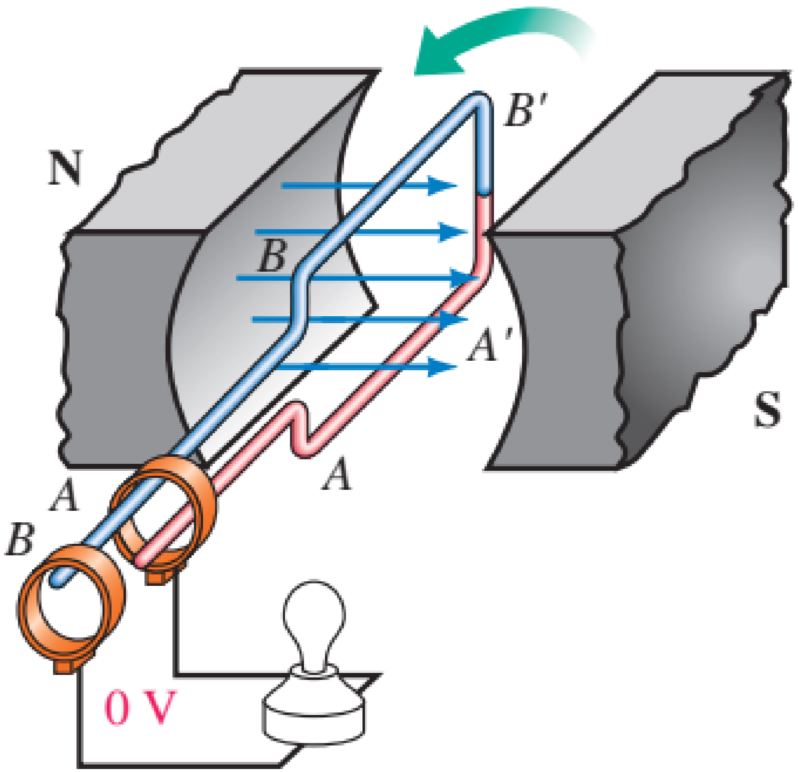


(a) 0° Position: Coil sides move parallel to flux lines. Since no flux is being cut, induced voltage is zero.

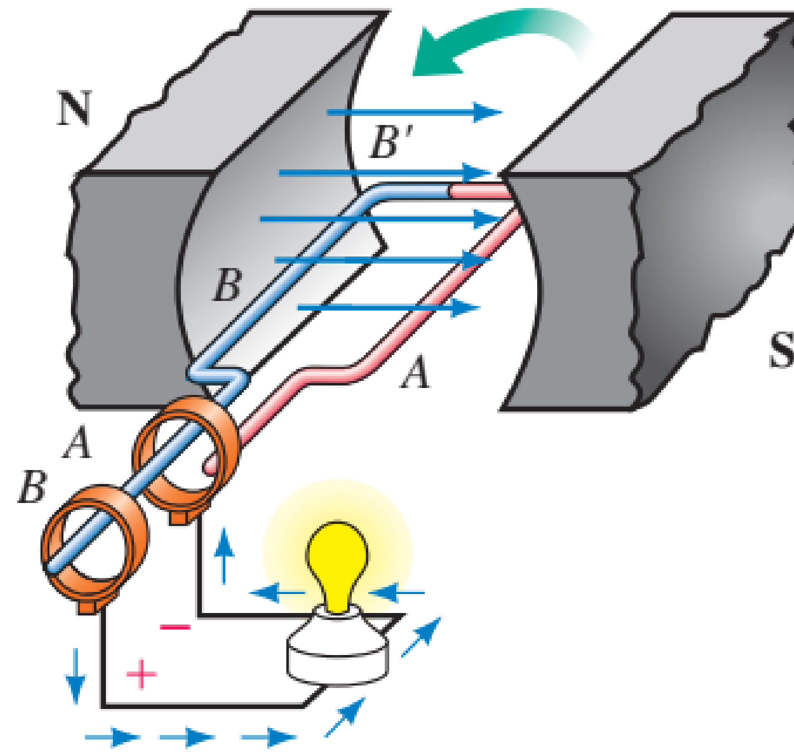


(b) 90° Position: Coil end *A* is positive with respect to *B*. Current direction is out of slip ring *A*.

4. Generating ac voltages (3/4)



(c) 180° Position: Coil again cutting no flux. Induced voltage is zero.

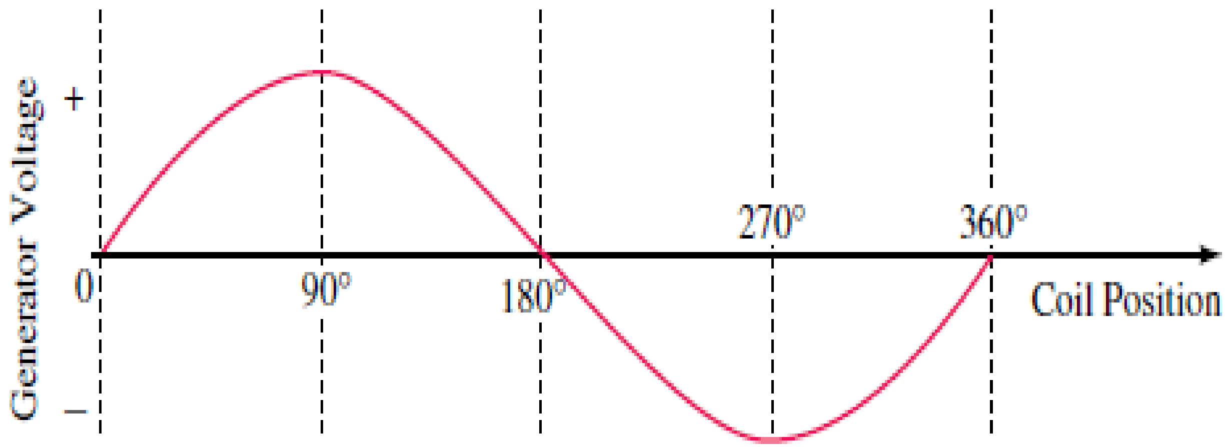


(d) 270° Position: Voltage polarity has reversed, therefore, current direction has also reversed.

4. Generating ac voltages (4/4)

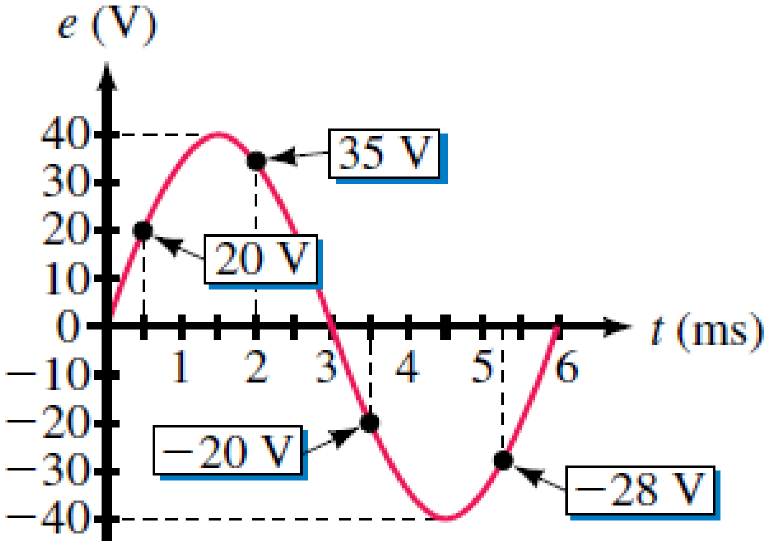
Remember!

- We called voltage a sinusoidal ac waveform.
- Therefore sin(0)= 0 induced voltage, sin(90) =+1 positive induced voltage, sin(180)=0 induced voltage and sin(270)= -1 negative induced voltage and so on.



5. Instantaneous value

- The value of waveform (voltage or current waveform) at any point is referred to as its ***instantaneous value***.
- For this example, the voltage waveform has a peak value of 40 volts and cycle time of 6ms. From graph we can see that at $t=0ms$, the voltage is zero. At $t=0.5ms$, it is 20V. At $t=2ms$, it is 35V. At $t=3.5ms$, it is -20V and so on.
- One cycle=One revolution.
- One cycle= 360°.



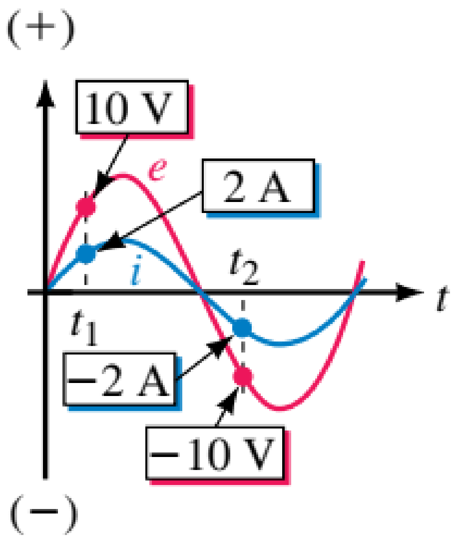
6. Voltage Convention for AC

- First we assign a ***reference polarity*** for source
- When voltage has a positive value
  - Its polarity is same as reference polarity
- When voltage is negative
  - Its polarity is opposite to that of the reference polarity

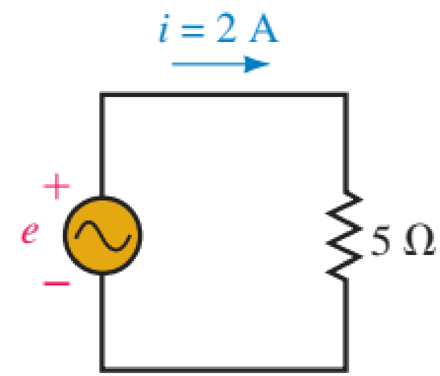
7. Current Convention for AC

- Assign a ***reference direction*** for current that leaves the source.
- When current has a positive value
  - Its actual direction is same as current reference arrow
- When current is negative
  - Its actual direction is opposite that of current reference arrow

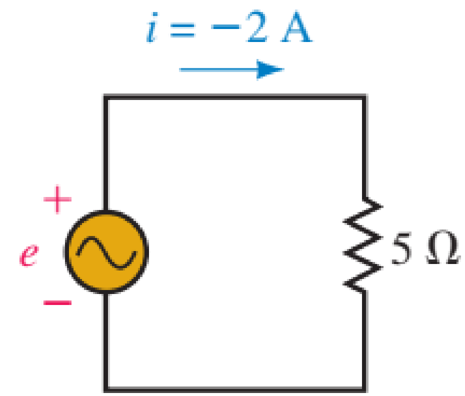
8. Illustrating AC Voltage and Current convention (1/3)



(a)



(b) Time t_1: $e = 10$ V and $i = 2$ A. Thus, voltage and current have the polarity and direction indicated.

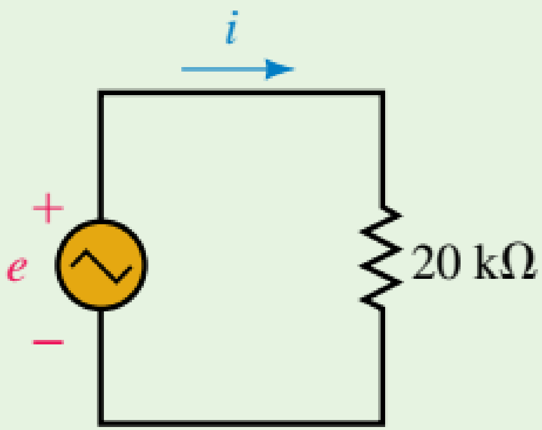


(c) Time t_2: $e = -10$ V and $i = -2$ A. Thus, voltage polarity is opposite to that indicated and current direction is opposite to the arrow direction.

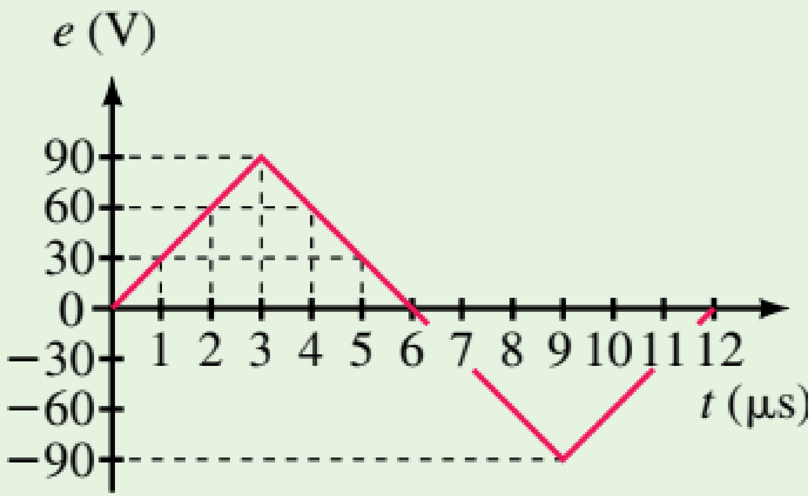
8. Illustrating AC Voltage and Current convention (2/3)

Example 1:

Figure below shows one cycle of a triangular voltage wave. Determine the current and its direction at t = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12 μs and sketch.



(a)

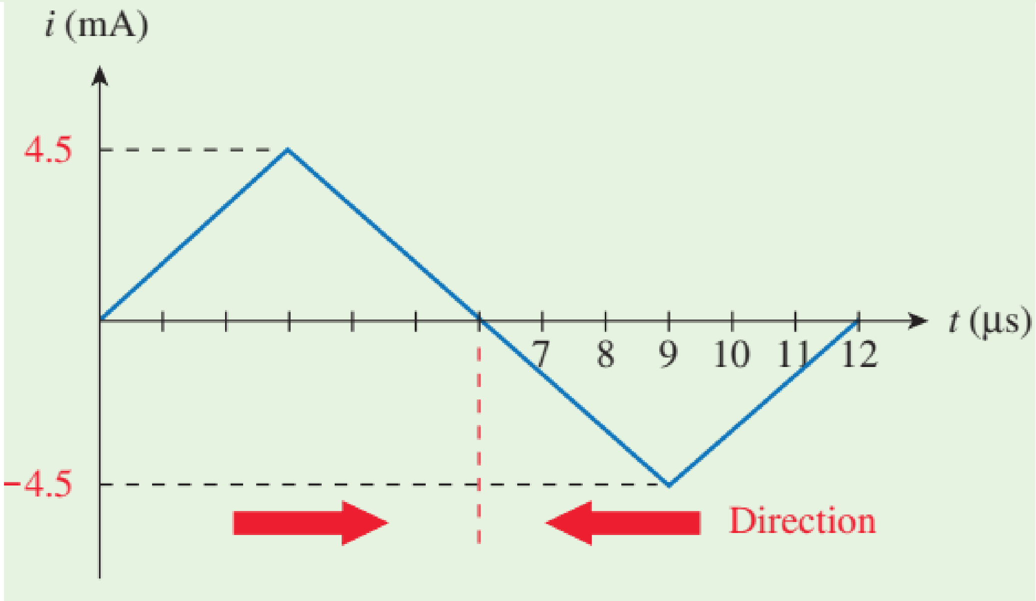


(b) Voltage

8. Illustrating AC Voltage and Current convention (3/3)

Example 1:

Solution Apply Ohm's law at each point in time. At $t = 0\ \mu s$, $e = 0$ V, so $i = e/R = 0\ V/20\ k\Omega = 0$ mA. At $t = 1\ \mu s$, $e = 30$ V. Thus, $i = e/R = 30\ V/20\ k\Omega = 1.5$ mA. At $t = 2\ \mu s$, $e = 60$ V. Thus, $i = e/R = 60\ V/20\ k\Omega = 3$ mA. Continuing in this manner, you get the values shown in Table 1. The waveform is plotted as Figure (c).



(c) Current

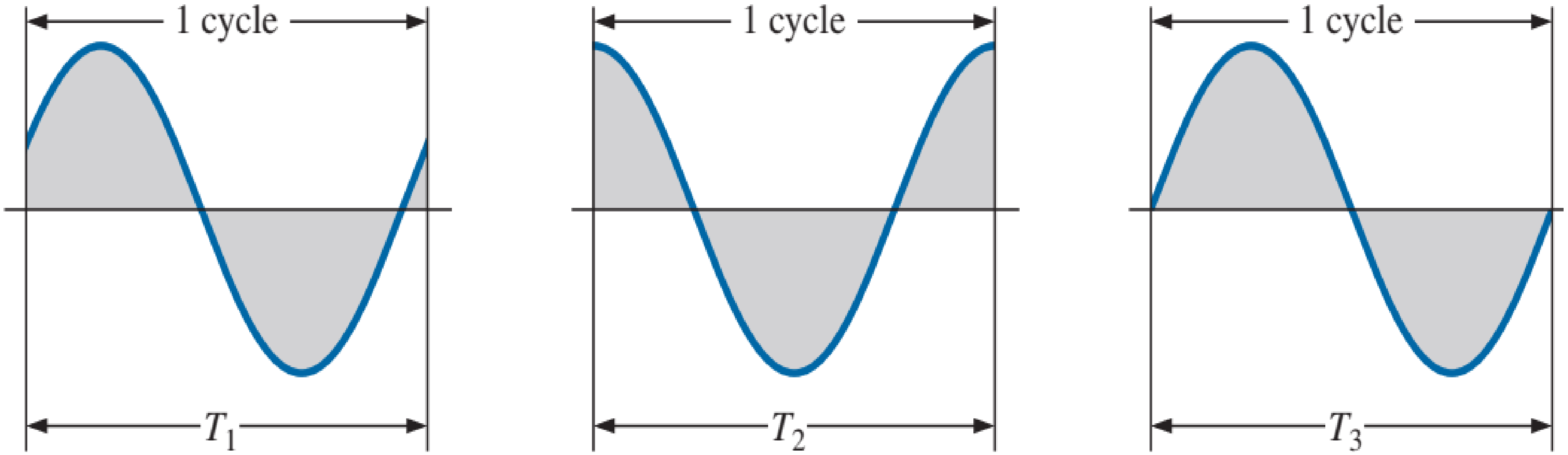
t (μs)	e (V)	i (mA)
0	0	0
1	30	1.5
2	60	3.0
3	90	4.5
4	60	3.0
5	30	1.5
6	0	0
7	−30	−1.5
8	−60	−3.0
9	−90	−4.5
10	−60	−3.0
11	−30	−1.5
12	0	0

Table 1

9. Frequency (1/3)

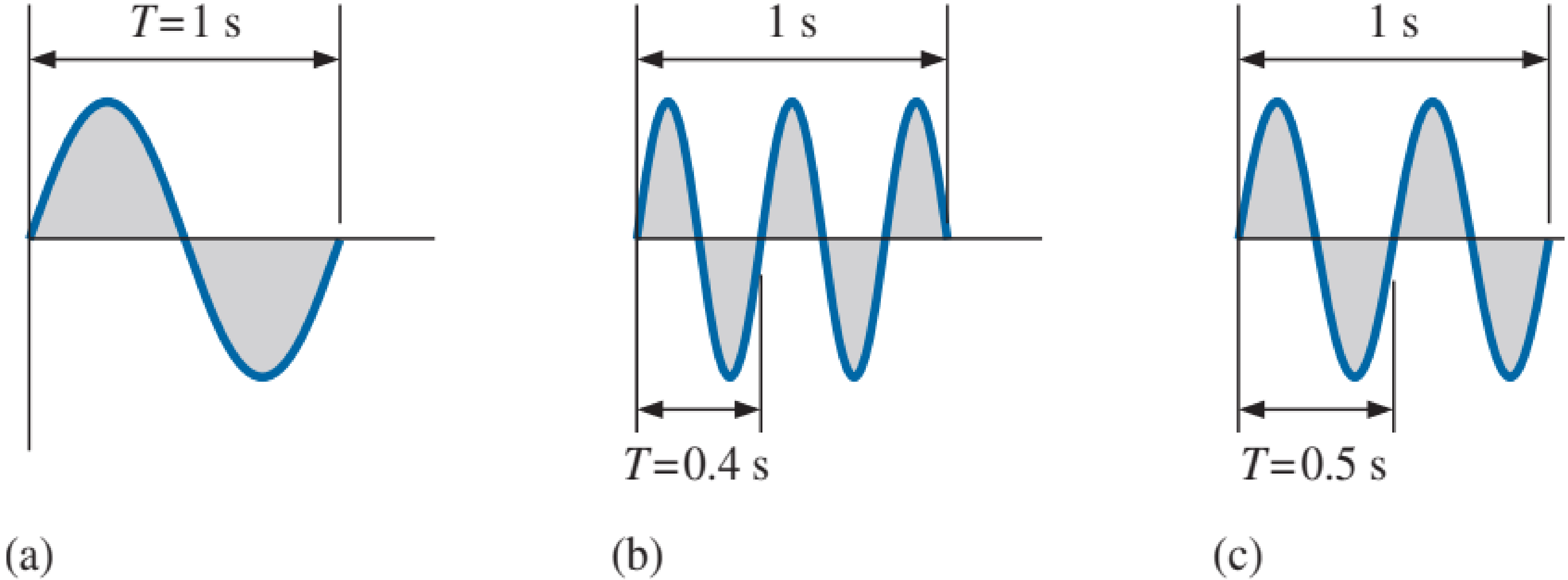
- Number of cycles that occur in 1 second.
  - Frequency
  - Denoted by f
- Unit of frequency is hertz (Hz)
- 1 Hz = 1 cycle per second

9. Frequency (2/3)



Defining the cycle and period of a sinusoidal waveform.

9. Frequency (3/3)



In (a), frequency, f is 1 cycle per second = 1 Hertz; T = 1/f = 1/1 = 1 second.
In (b), frequency, f is 2.5 cycles per second = 2.5 Hertz; T = 1/f = 1/2.5 = 0.4 second.
In (c), frequency, f is 2 cycles per second = 2 Hertz ; T = 1/f = 1/2 = 0.5 second.

10. Period

- Period of a waveform
  - Duration of one cycle.
- Time is measured in seconds
- The period is the reciprocal of frequency
  - $T = 1/f$ *(seconds)*

11. Definitions (1/4)

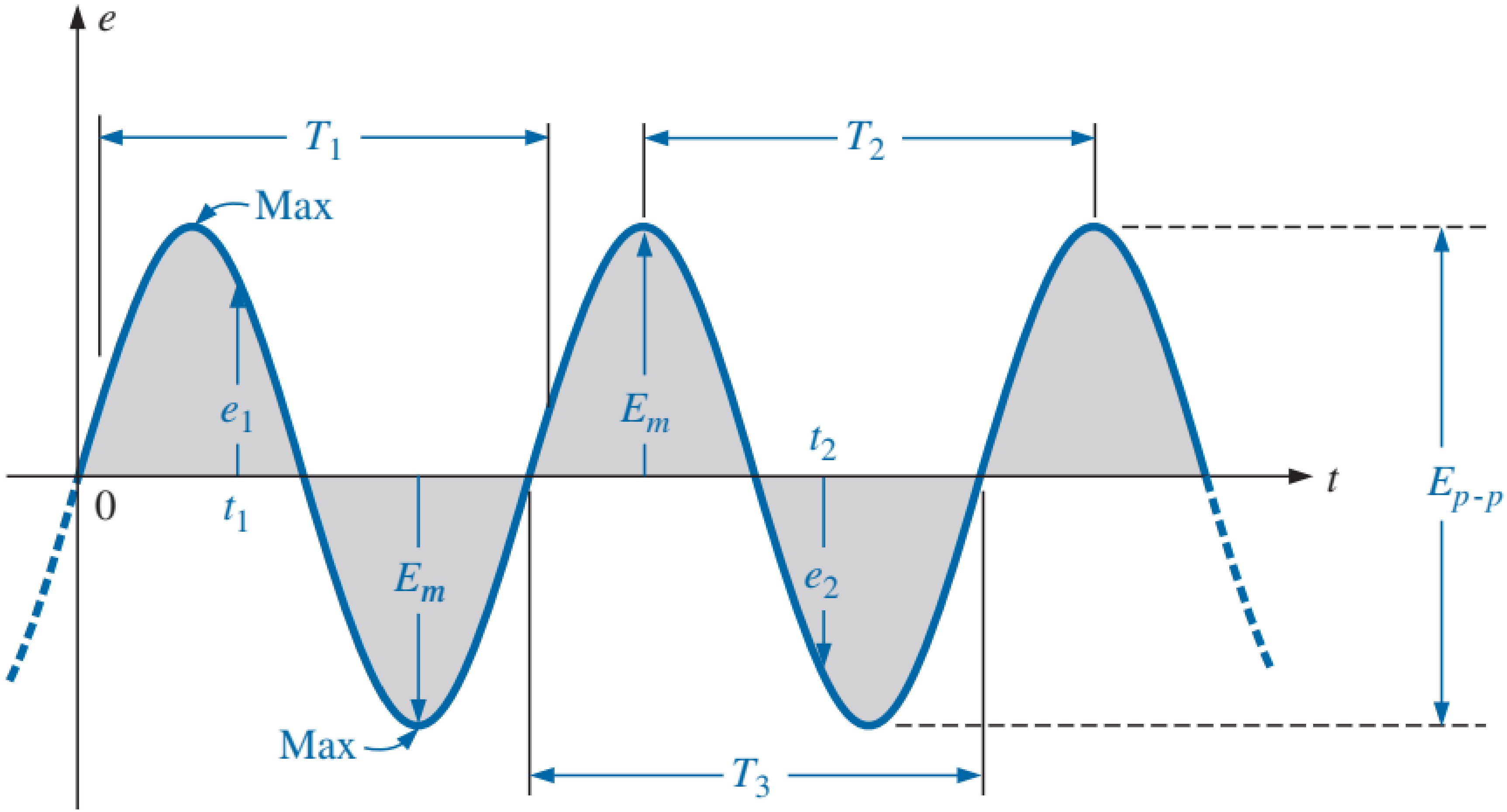
Instantaneous value: The magnitude of a waveform at any instant of time; denoted by lowercase letters (e_1, e_2 in Figure next slide).

Peak amplitude: The maximum value of a waveform as measured from its *average*, or *mean*, value, denoted by uppercase letters [such as E_m . For the waveform in Figure next slide, the average value is zero volts, and E_m is as defined by the figure.

Peak value: The maximum instantaneous value of a function as measured from the zero volt level. For the waveform in Figure, thepeak amplitude and peak value are the same since the average valueof the function is zero volts.

Peak-to-peak value: Denoted by $E_{p\text{-}p}$, the full voltage between positive and negative peaks of the waveform, that is, the sum of the magnitude of the positive and negative peaks.

11. Definitions (2/4)



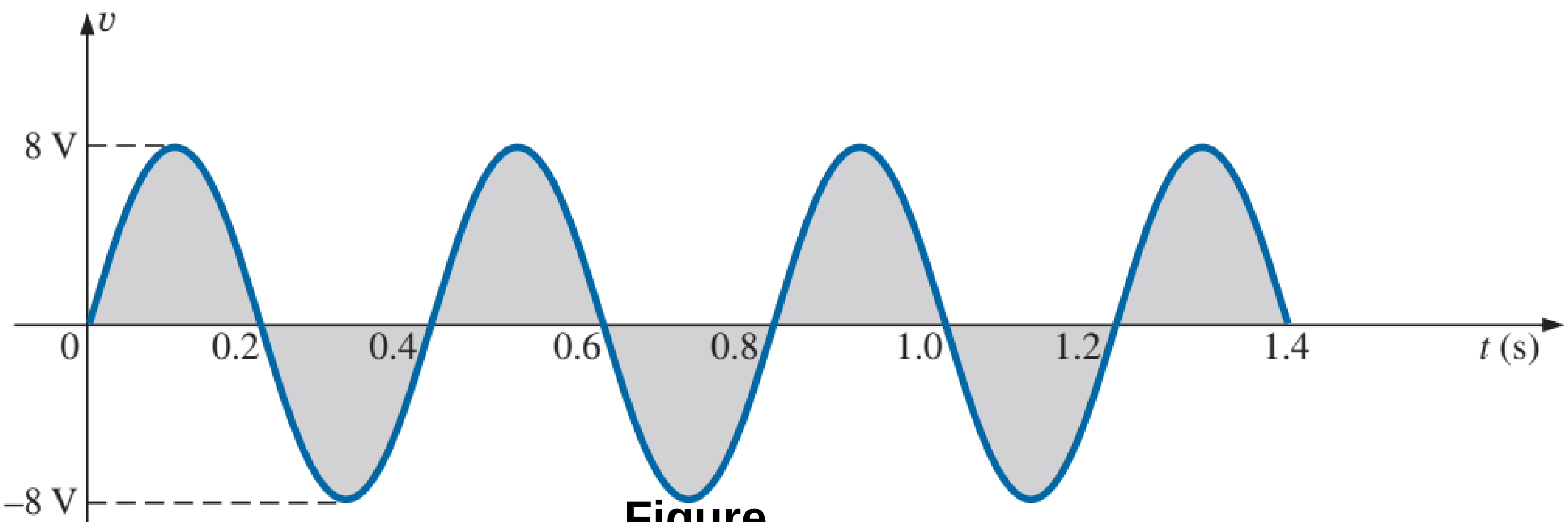
FIGURE

Important parameters for a sinusoidal voltage.

11. Definitions (3/4)

EXAMPLE 2 For the sinusoidal waveform in Figure below:

a. What is the peak value?
b. What is the instantaneous value at 0.3 s and 0.6 s?
c. What is the peak-to-peak value of the waveform?
d. What is the period of the waveform?
e. How many cycles are shown?
f. What is the frequency of the waveform?



Figure

11. Definitions (4/4)

Solutions:

a. **8 V.**

b. At 0.3 s, **−8 V;** at 0.6 s, **0 V.**

c. **16 V.**

d. **0.4 s.**

e. **3.5 cycles.**

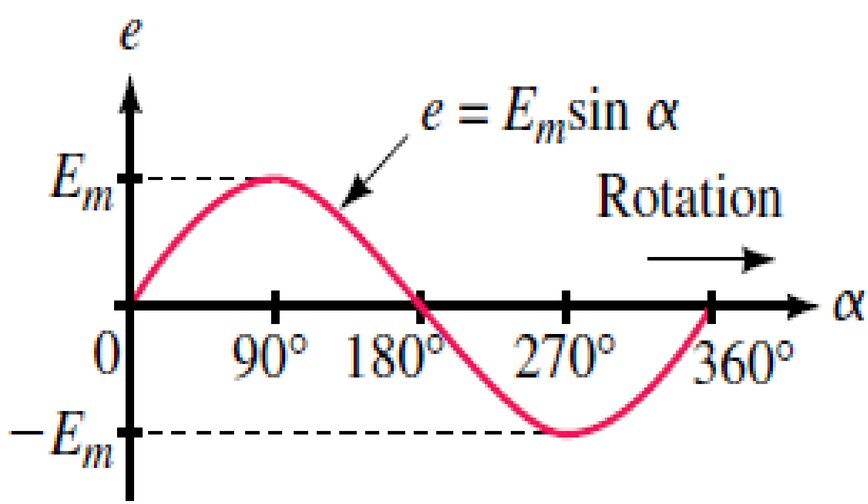
f. **2.5 cps,** or **2.5 Hz.** (f =1/T = 1/0.4 s = 2.5 Hz)

12. The Basic Sine Wave Equation (1/3)

- Voltage produced by a generator is (Recall slides 7 and 8)
  - $e = E_m \sin \alpha$ (V)   …   (i)
- E_m is the peak voltage or maximum coil voltage.
- α is instantaneous angular position of rotating coil of the generator.

12. The Basic Sine Wave Equation (2/3)

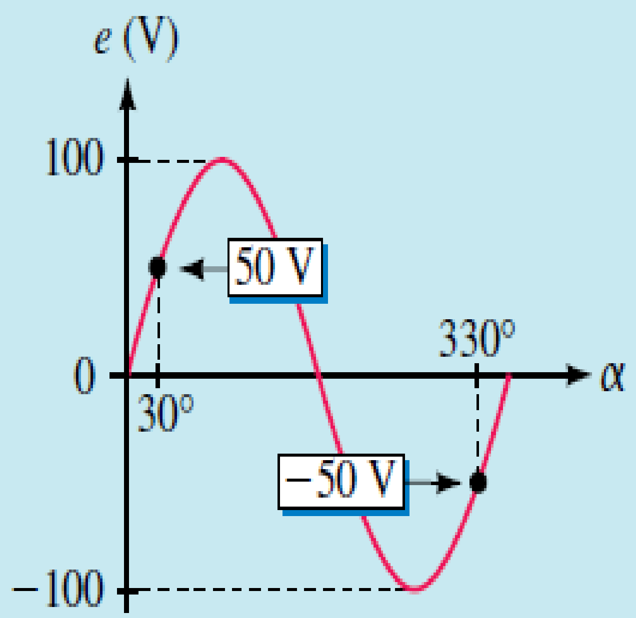
- Equation (i) states that the voltage at any point on the sine wave may be found by multiplying E_m times the sine of the angle at that point.



12. The Basic Sine Wave Equation (3/3)

- **Example 3:** If the amplitude of the waveform of figure below is $E_m = 100\,V$, Determine the coil voltage at 30°and 330°

Solution At $\alpha = 30°$, $e = E_m \sin \alpha = 100 \sin 30° = 50$ V. $e = 100 \sin 330° = -50$ V. These are shown on the graph of Figure Below:



α

13. Angular Velocity (1/2)

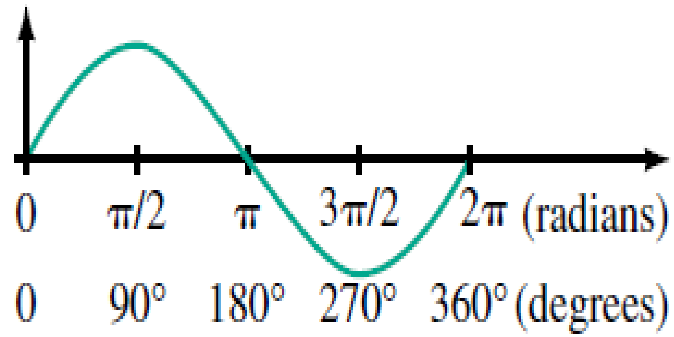
- Rate at which the generator coil rotates with respect to time, ω (Greek letter omega).
- If the coil rotates through an angle of 30° in one second. For example, its angular velocity is 30° per second.
- For the case cited, $\omega = 30°/s$.
- Normally angular velocity is expressed in radians per seconds instead of degrees per second.
  - $\omega = \alpha / t$
  - $t = \alpha / \omega$ (s)
  - $\alpha = \omega t$

13. Angular Velocity (2/2)

- ω is usually expressed in radians/second
- 2π radians = 360°
- To convert from degrees to radians, multiply by $\pi/180$
- To convert from radians to degrees, multiply by $180/\pi$

$$\alpha_{\text{radians}} = \frac{\pi}{180^\circ} \times \alpha_{\text{degrees}}$$

$$\alpha_{\text{degrees}} = \frac{180^\circ}{\pi} \times \alpha_{\text{radians}}$$



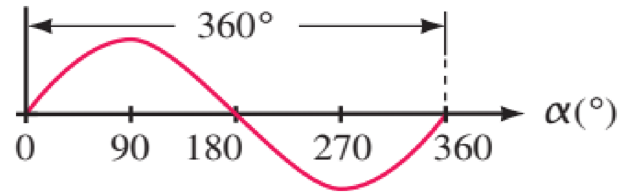
$$\mathbf{90^\circ}\textbf{:}\ \text{Radians} = \frac{\pi}{180^\circ}(90^\circ) = \frac{\boldsymbol{\pi}}{\mathbf{2}}\ \textbf{rad}$$

$$\mathbf{30^\circ}\textbf{:}\ \text{Radians} = \frac{\pi}{180^\circ}(30^\circ) = \frac{\boldsymbol{\pi}}{\mathbf{6}}\ \textbf{rad}$$

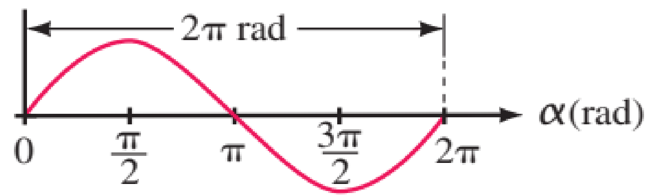
$$\frac{\mathbf{3}\boldsymbol{\pi}}{\mathbf{2}}\ \textbf{rad:}\ \text{Degrees} = \frac{180^\circ}{\pi}\left(\frac{3\pi}{2}\right) = \mathbf{270^\circ}$$

14. Relationship between ω, T, and f

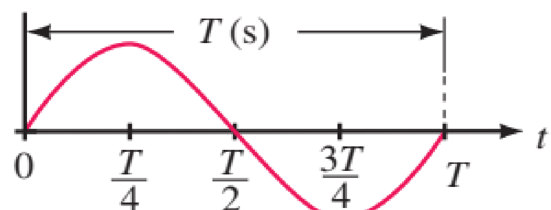
- Earlier we learned that one cycle of a sine wave may be represented as either $\alpha = 2\pi$ rads or $t = T$ sec.



(a) Degrees



(b) Radians



(c) Period

- $\alpha = \omega T$
- $2\pi = \omega T$ (rad)
- $\omega = 2\pi / T$ (rad/sec)
- $\omega = 2\pi \times 1/T$ (rad/sec)
- $\omega = 2\pi \times f$ (rad/sec)
- $\omega = 2\pi f$ (rad/sec)

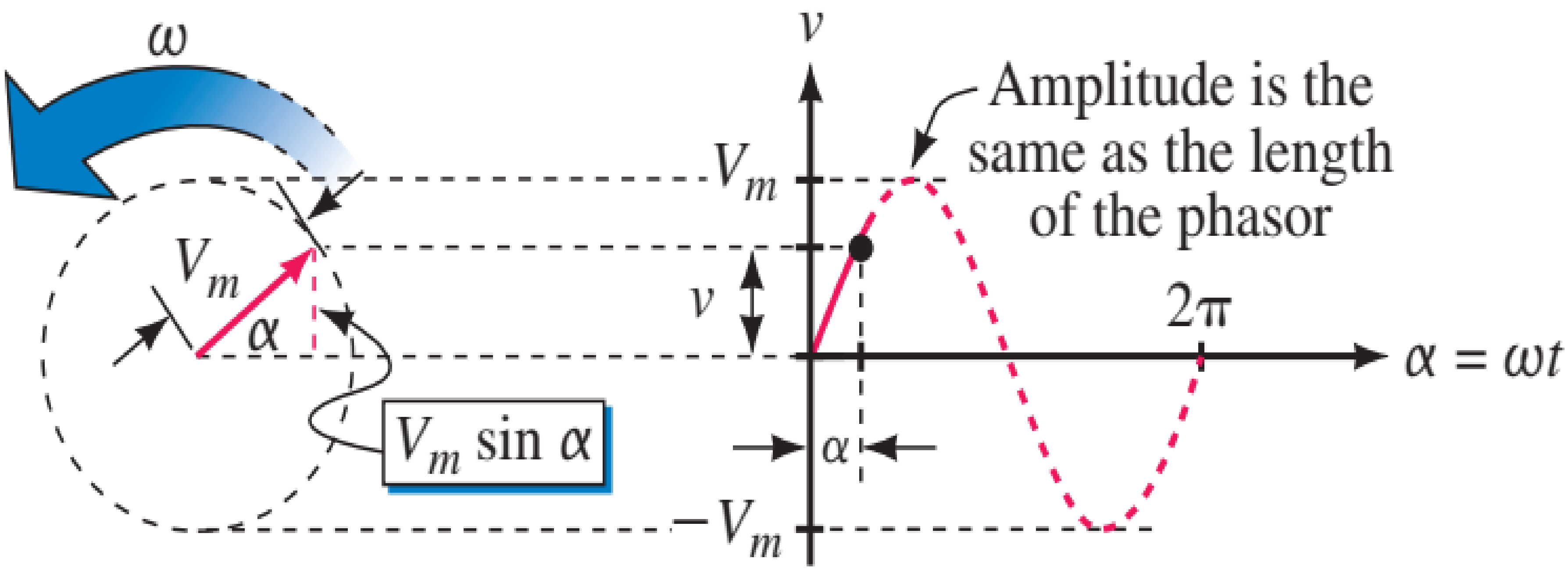
15. Voltages and Currents as Functions of Time

- Since $\alpha = \omega t$, the equation $e = E_m \sin \alpha$ becomes $e(t) = E_m \sin \omega t$
- Also, $v(t) = V_m \sin \omega t$ and $i(t) = I_m \sin \omega t$

16. Phasors (1/6)

- A phasor is a rotating line whose projection on a vertical axis can be used to represent sinusoidally varying quantities.
- Consider the red line of length V_m shown in (figure (a) next slide). It is called a phasor. The vertical projection of this line (indicated in dotted red) is $V_m \sin\alpha$.
- Now assume that the phasor rotates at angular velocity of ω rad/sec in the anticlockwise direction .Then $\alpha = \omega t$ and its vertical projection is $V_m \sin\omega t$.
- If we designate this projection (height) as v, we get $v = V_m \sin\omega t$. Which is the familiar sinusoidal voltage equation.
- If we plot a graph of v versus α , we get the sine wave as shown in (figure (b) next slide)

16. Phasors (2/6)



(a) Phasor

(b) Resulting sine wave

16. Phasors (3/6)

- Consider the figure (c) next slide(s) illustrating the process which shows the snapshots of the phasor and the evolving waveform at various instants of time.
- Consider phasor magnitude (Phasor length) V_m =100V rotating at ω=30°/s. For example consider t = 0, 1, 2 and 3s Suppose for t = 3 sec, $v = V_m \sin\omega t = v = 100\sin(30°)(3) = 100\text{V}$
- ***Phasors apply only to sinusoidal waveforms***.
- If the phasor has a length of V_m , it represents voltage; if the phasor has a length of I_m , it represents current.

16. Phasors (4/6)

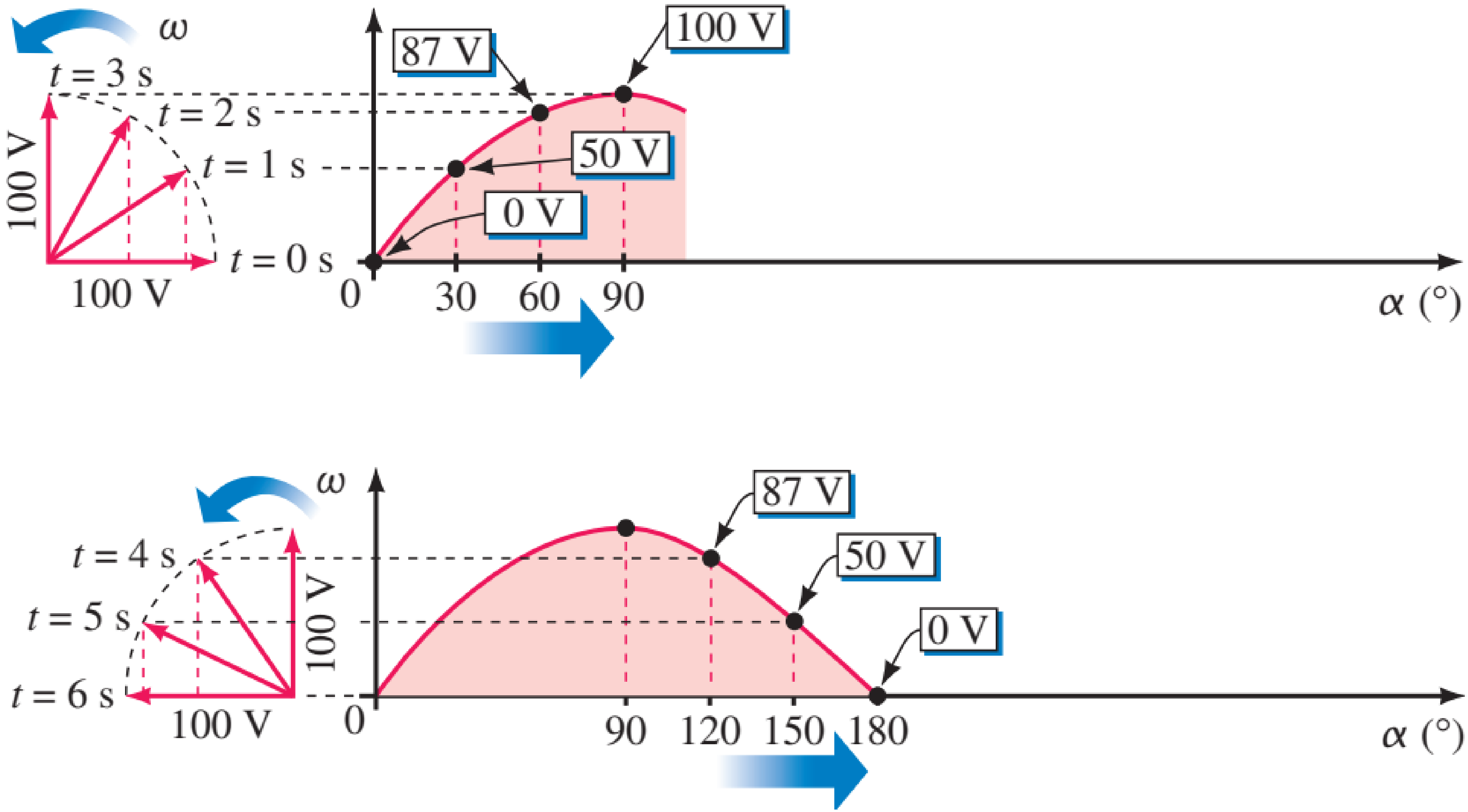


Figure (c)

16. Phasors (5/6)

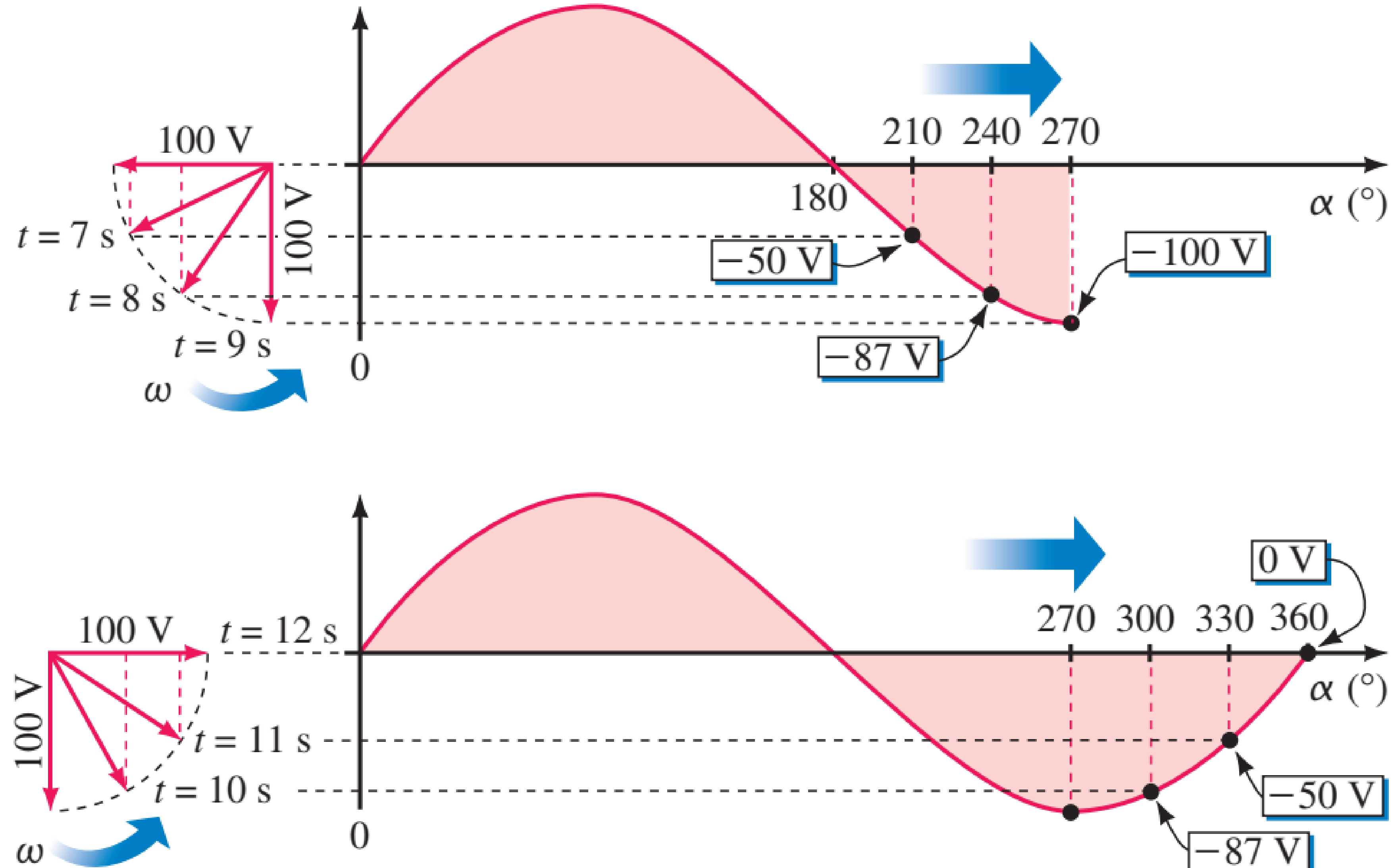
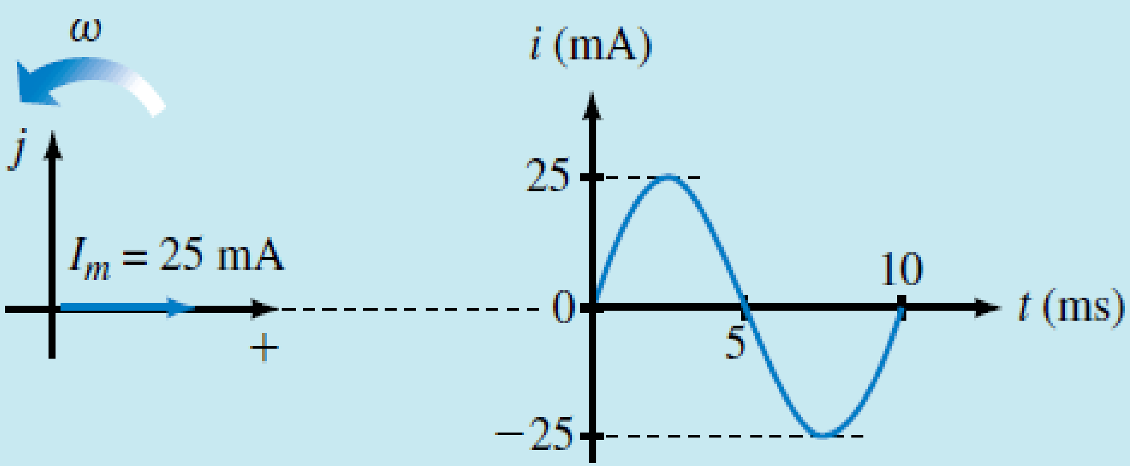


Figure (c) ***continued…***

16. Phasors (6/6)

EXAMPLE 4 Draw the phasor and waveform for current $i = 25 \sin \omega t$ mA for $f = 100$ Hz.

Solution The phasor has a length of 25 mA and is drawn at its $t = 0$ position, which is zero degrees as indicated in Figure . Since $f = 100$ Hz, the period is $T = 1/f = 10$ ms.



17. Phase Relations (1/9)

Thus far, we have considered only sine waves that have maxima at $\pi/2$ and $3\pi/2$, with a zero value at 0, π, and 2π, as shown in Fig. 1. If the waveform is shifted to the right or left of 0°, the expression becomes

$$\boxed{A_m \sin(\omega t \pm \theta)}$$

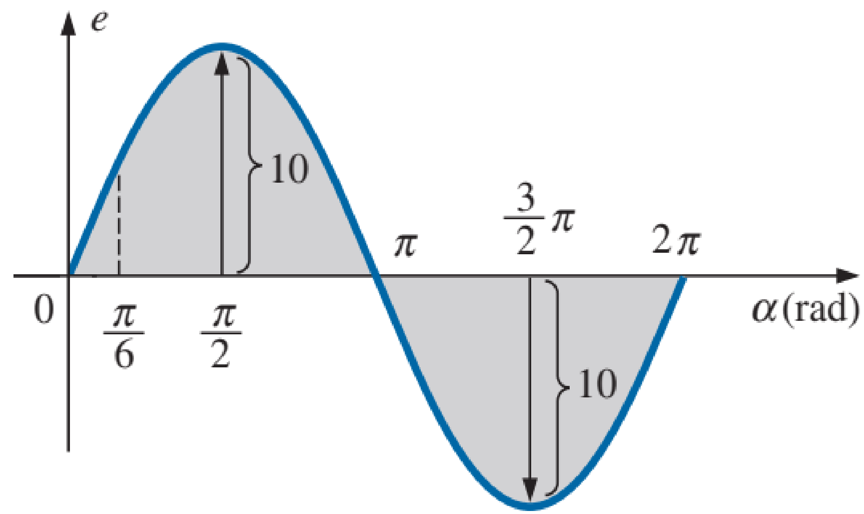


FIG. 1

horizontal axis in radians.

17. Phase Relations (2/9)

where θ is the angle in degrees or radians that the waveform has beenshifted. If the waveform passes through the horizontal axis with a *positive-going* (increasing with time) slope *before* 0°, as shown in Fig. 2, theexpression is

$$\boxed{A_m \sin(\omega t + \theta)}$$

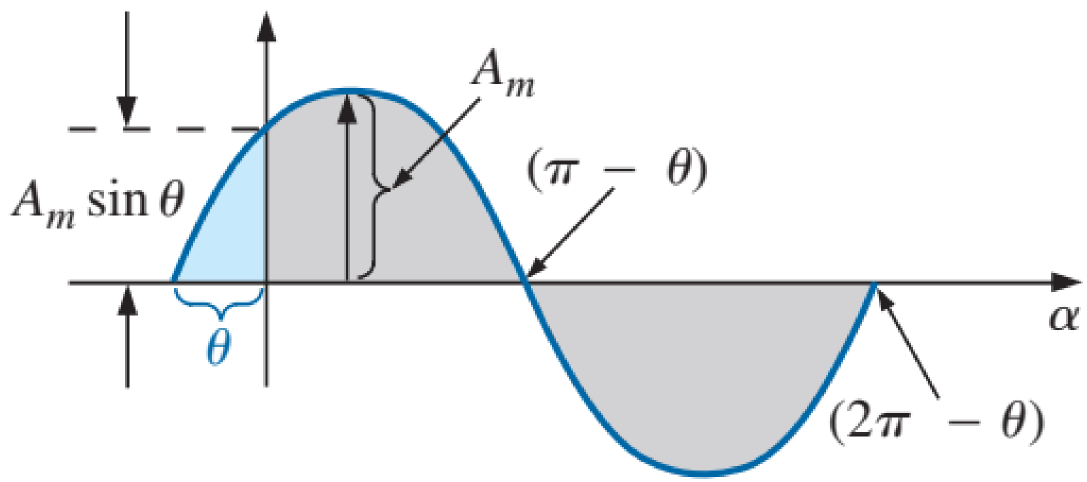


FIG. 2

17. Phase Relations (3/9)

At $\omega t = \alpha = 0°$, the magnitude is determined by $A_m \sin\theta$. If the waveform passes through the horizontal axis with a positive-going slope *after* 0°, as shown in Fig. 3, the expression is

$$\boxed{A_m \sin(\omega t - \theta)}$$

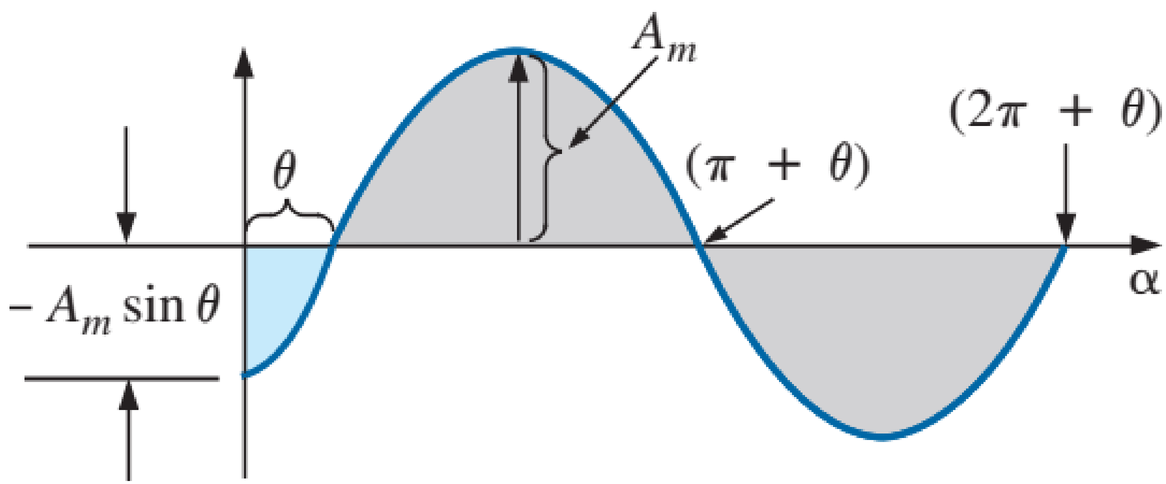


FIG. 3

α

17. Phase Relations (4/9)

Finally, at $\omega t = \alpha = 0°$, the magnitude is $A_m \sin(-\theta)$, which, by a trigonometric identity, is $-A_m \sin\theta$.

If the waveform crosses the horizontal axis with a positive-going slope 90° ($\pi/2$) sooner, as shown in Fig. 4, it is called a *cosine wave;* that is,

$$\boxed{\sin(\omega t + 90°) = \sin\left(\omega t + \frac{\pi}{2}\right) = \cos \omega t}$$

or

$$\boxed{\sin \omega t = \cos(\omega t - 90°) = \cos\left(\omega t - \frac{\pi}{2}\right)}$$

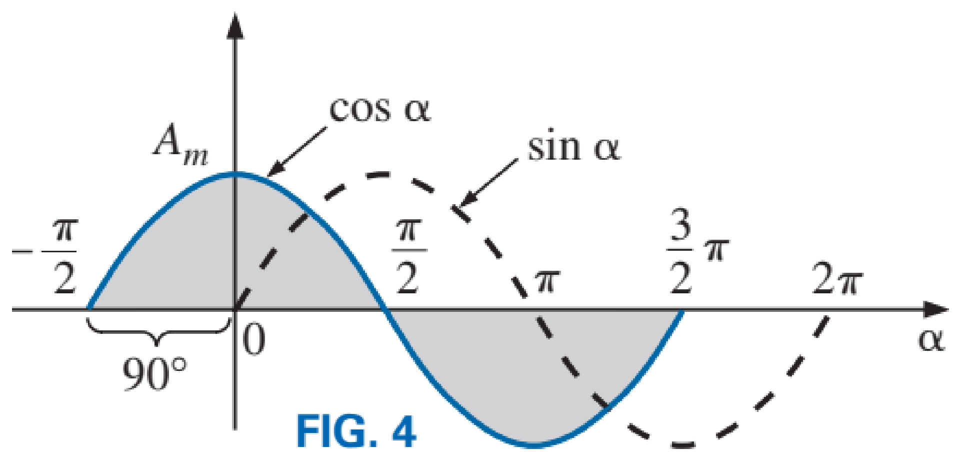


FIG. 4

α

17. Phase Relations (5/9)

The terms **leading** and **lagging** are used to indicate the relationship between two sinusoidal waveforms of the *same frequency* plotted on the same set of axes. In Fig. 4, the cosine curve is said to *lead* the sine curve by 90°, and the sine curve is said to *lag* the cosine curve by 90°. The 90° is referred to as the phase angle between the two waveforms. In language commonly applied, the waveforms are *out of phase* by 90°.

If both waveforms cross the axis at the same pointwith the same slope, they are *in phase*.

If a sinusoidal expression appears as

$$e = -E_m \sin \omega t = E_m(-\sin \omega t)$$

$$\cos \alpha = \sin(\alpha + 90°)$$
$$\sin \alpha = \cos(\alpha - 90°)$$

$$\sin(-\alpha) = -\sin \alpha$$
$$\cos(-\alpha) = \cos \alpha$$

17. Phase Relations (6/9)

EXAMPLE 5 What is the phase relationship between the sinusoidal waveforms of each of the following sets?

a. $v = 10 \sin(\omega t + 30°)$
   $i = 5 \sin(\omega t + 70°)$

b. $i = 15 \sin(\omega t + 60°)$
   $v = 10 \sin(\omega t - 20°)$

c. $i = 2 \cos(\omega t + 10°)$
   $v = 3 \sin(\omega t - 10°)$

17. Phase Relations (7/9)

Solutions:

a. See Fig. 5.
***i* leads *v* by 40º, or *v* lags *i* by 40º.**

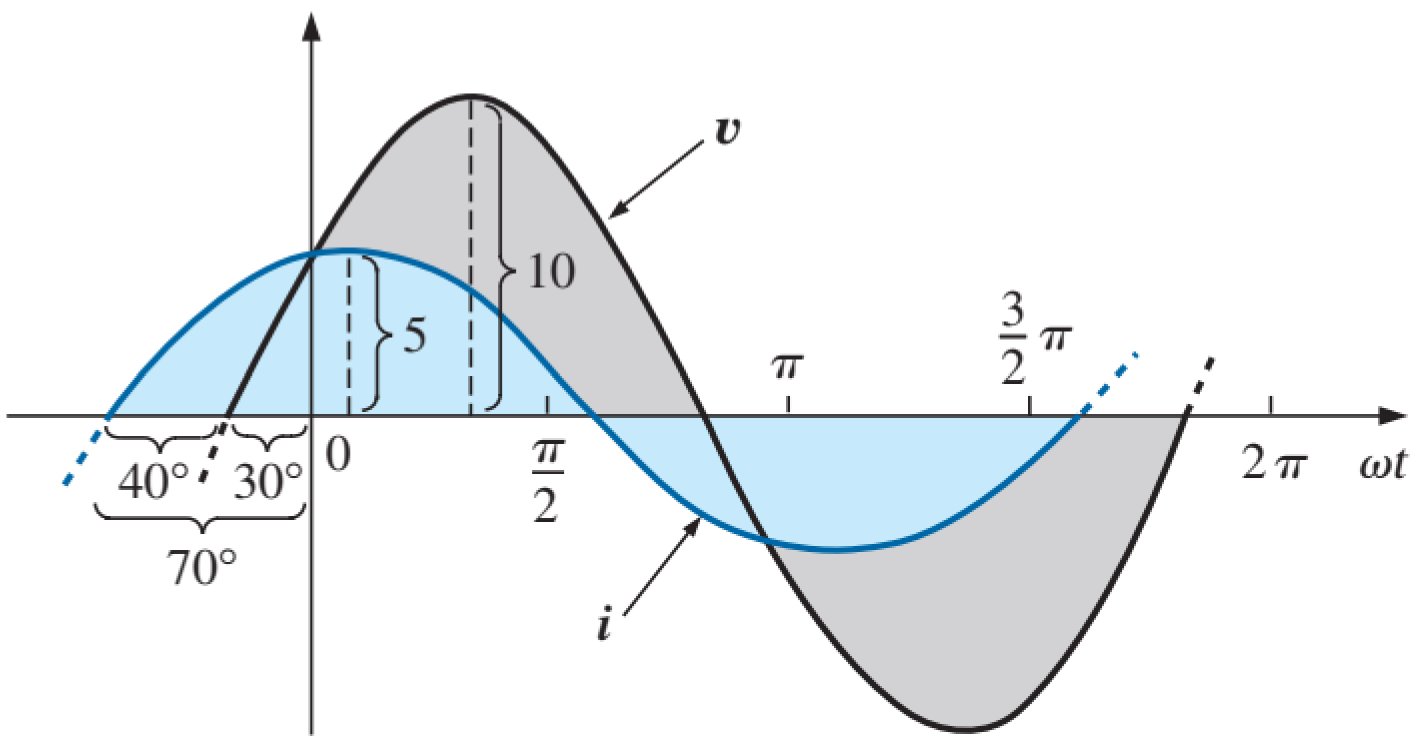


FIG. 5

17. Phase Relations (8/9)

b. see Fig. 6.

i leads v by 80º, or v lags i by 80º.

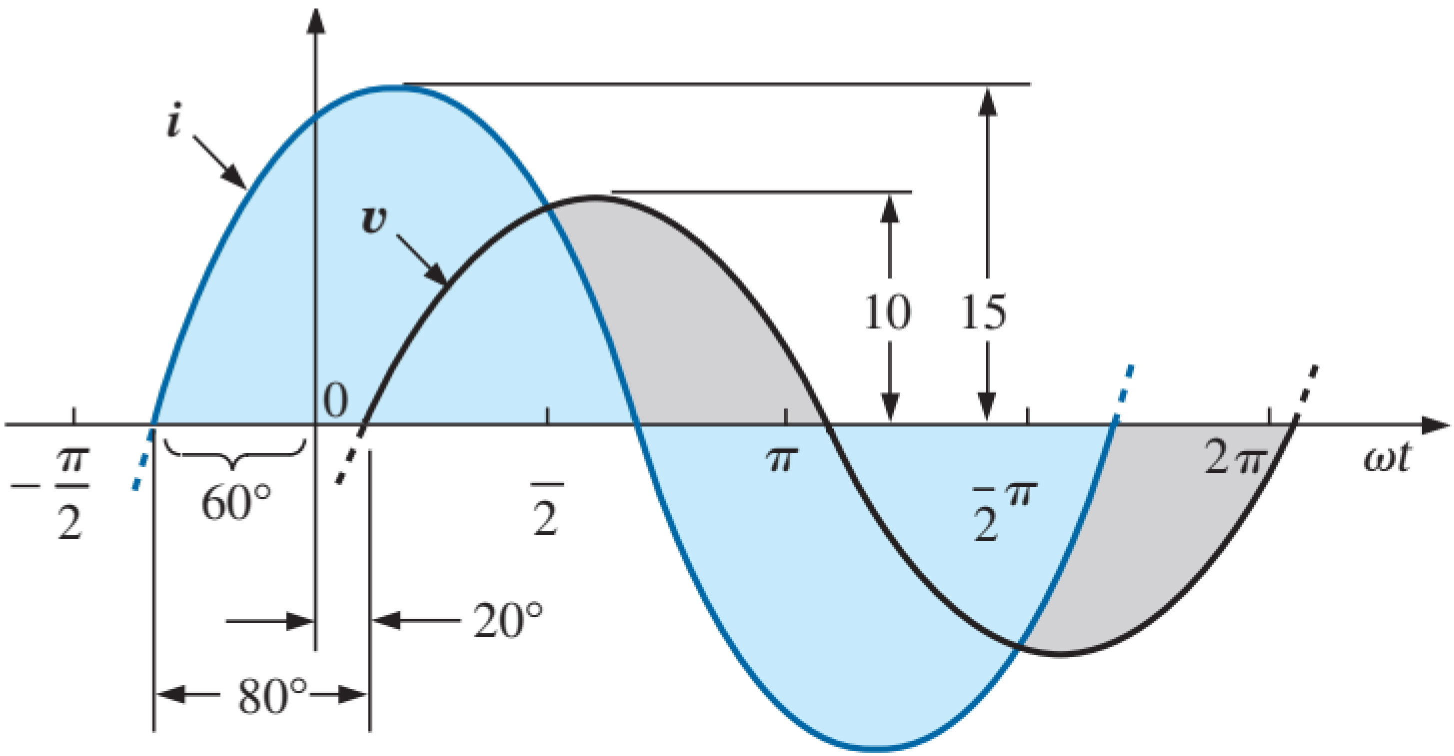


FIG. 6

17. Phase Relations (9/9)

c. See Fig. 7.

$$i = 2\cos(\omega t + 10°) = 2\sin(\omega t + 10° + 90°)$$
$$= 2\sin(\omega t + 100°)$$

***i* leads *v* by 110º, or *v* lags *i* by 110º.**

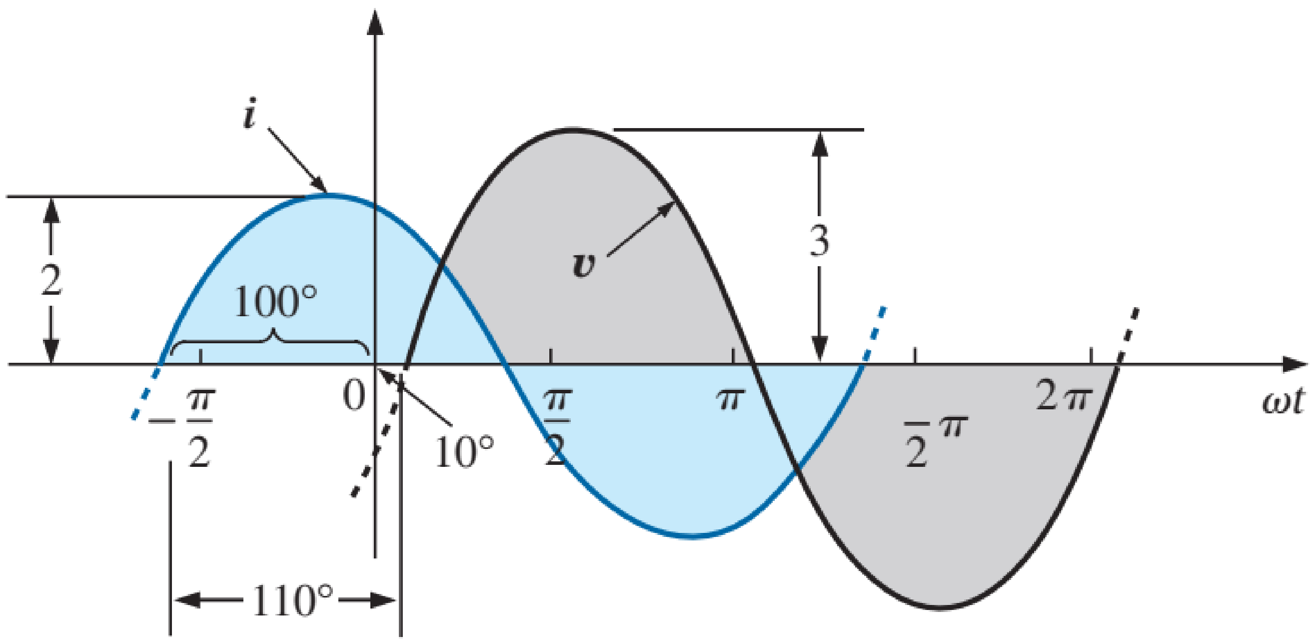


FIG. 7

18. Average Value (1/12)

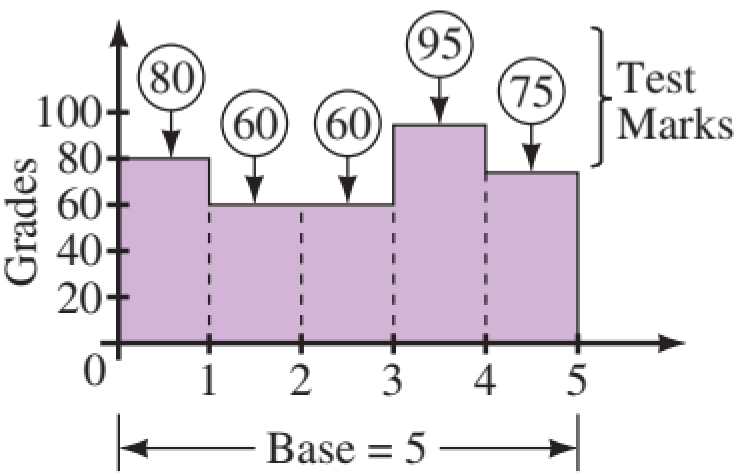
- To find an average value of a waveform
  - Average = area under curve/ length of base
- Areas above axis are positive, areas below axis are negative
- Average values are also called *dc* values, because dc meters indicate average values rather than instantaneous values.

18. Average Value (2/12)

Average in Terms of the Area under a Curve

An approach more suitable for use with waveforms is to find the area under the curve, then divide by the baseline of the curve. To get at the idea, we can use an analogy. Consider again the technique of computing the average for a set of numbers. Assume that you earn marks of 80, 60, 60, 95, and 75 on a group of tests. Your average mark is therefore

$$\text{average} = (80 + 60 + 60 + 95 + 75)/5 = 74$$



18. Average Value (3/12)

An alternate way to view these marks is graphically as in Figure previous slide.The area under this curve can be computed as

$$\text{area} = (80 \times 1) + (60 \times 2) + (95 \times 1) + (75 \times 1)$$

Now divide this by the length of the base, namely 5. Thus,

$$\frac{(80 \times 1) + (60 \times 2) + (95 \times 1) + (75 \times 1)}{5} = 74$$

which is exactly the answer obtained in previous slide. That is,

$$\text{average} = \frac{\text{area under curve}}{\text{length of base}}$$

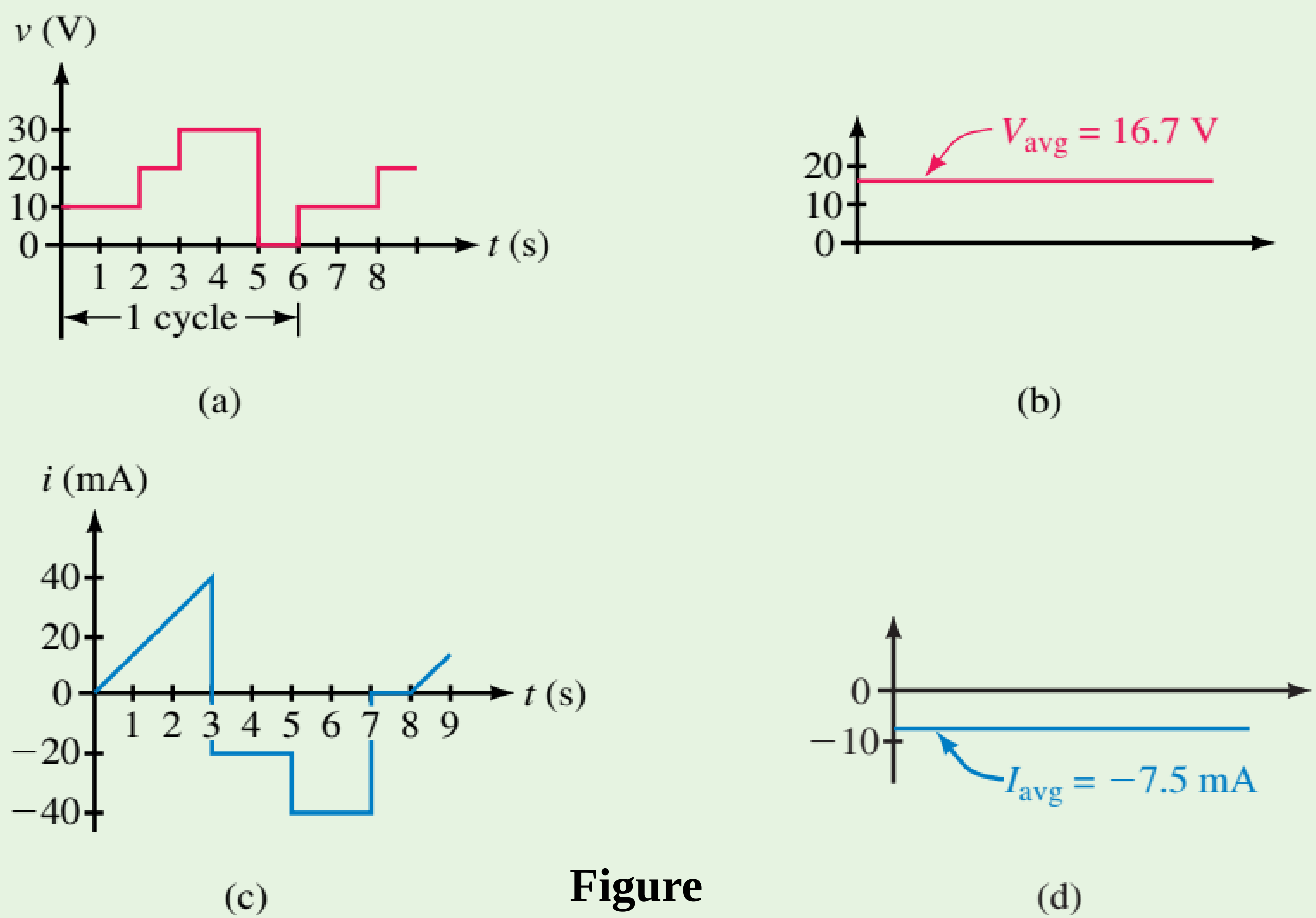
or

$$\boxed{\text{Average}(G) = \frac{\text{area under curve}}{\text{length of curve}}}$$

18. Average Value (4/12)

Example 6

Compute the average value for the waveforms of Figures (a) and (c). Sketch the averages for each.



Figure

18. Average Value (5/12)

Example 6

Solution For the waveform of (a), $T = 6$ s. Thus,

$$V_{avg} = \frac{(10\,\text{V} \times 2\,\text{s}) + (20\,\text{V} \times 1\,\text{s}) + (30\,\text{V} \times 2\,\text{s}) + (0\,\text{V} \times 1\,\text{s})}{6\,\text{s}} = \frac{100\,\text{V-s}}{6\,\text{s}} = 16.7\,\text{V}$$

The average is shown as (b). For the waveform of (c), $T = 8$ s and

$$I_{avg} = \frac{\frac{1}{2}(40\,\text{mA} \times 3\,\text{s}) - (20\,\text{mA} \times 2\,\text{s}) - (40\,\text{mA} \times 2\,\text{s})}{8\,\text{s}} = \frac{-60}{8}\,\text{mA} = -7.5\,\text{mA}$$

18. Average Value (6/12)

We found the areas under the curves in Example 6 by using a simple geometric formula. If we should encounter a sine wave or any other unusual shape, however, we must find the area by some other means. We can obtain a good approximation of the area by attempting to reproduce the original wave shape using a number of small rectangles or other familiar shapes, the area of which we already know through simple geometric formulas. For example,

the area of the positive (or negative) pulse of a sine wave is $2A_m$.

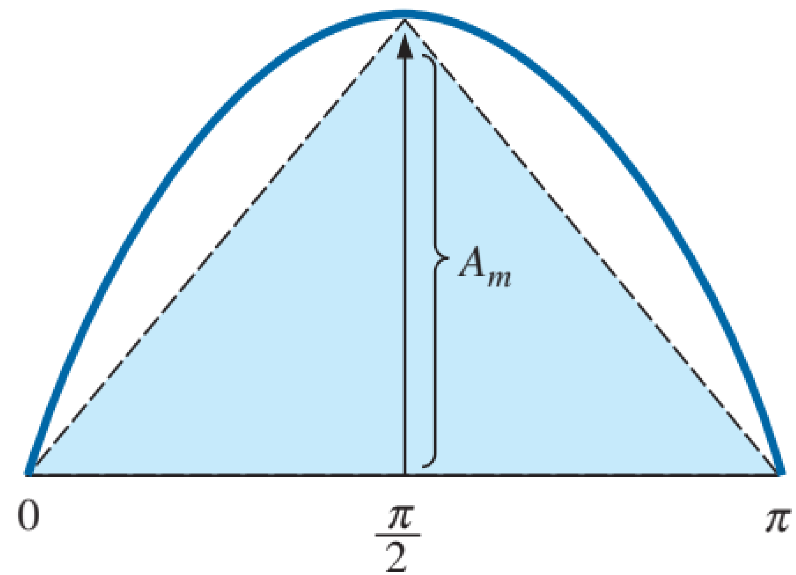
18. Average Value (7/12)

Approximating this waveform by two triangles (Fig. 8), we obtain (using *area = 1/2 base × height* for the area of a triangle) a rough idea of the actual area:

$$\text{Area shaded} = 2\left(\frac{1}{2}bh\right) = 2\left[\left(\frac{1}{2}\right)\overbrace{\left(\frac{\pi}{2}\right)}^{b}\overbrace{(A_m)}^{h}\right] = \frac{\pi}{2}A_m \cong 1.58A_m$$

FIG. 8

Approximating the shape of the positive pulse of a sinusoidal waveform with two right triangles.



18. Average Value (8/12)

A closer approximation may be a rectangle with two similar triangles (Fig. 9):

$$\text{Area} = A_m\frac{\pi}{3} + 2\left(\frac{1}{2}bh\right) = A_m\frac{\pi}{3} + \frac{\pi}{3}A_m = \frac{2}{3}\pi A_m = 2.094A_m$$

which is certainly close to the actual area. For irregular waveforms, this method can be especially useful if data such as the average value are desired.

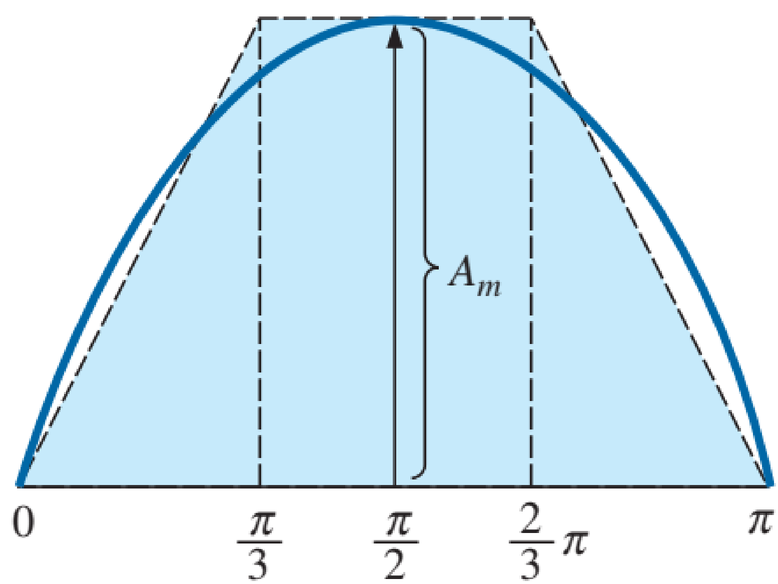


FIG. 9

A better approximation for the shape of the positive pulse of a sinusoidal waveform.

18. Average Value (9/12)

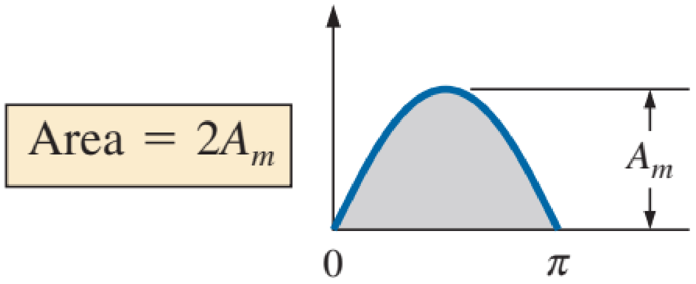
The procedure of calculus that gives the exact solution $2A_m$ is known as *integration*. Finding the area under the positive pulse of a sine wave using integration, we have

$$\text{Area} = \int_0^{\pi} A_m \sin \alpha \, d\alpha$$

where $A_m \sin \alpha$ is the function to be integrated, and $d\alpha$ indicates that we are integrating with respect to α.

Integrating, we obtain

$$\begin{aligned}\text{Area} &= A_m[-\cos \alpha]_0^{\pi}\\ &= -A_m(\cos \pi - \cos 0^\circ)\\ &= -A_m[-1 - (+1)] = -A_m(-2)\end{aligned}$$



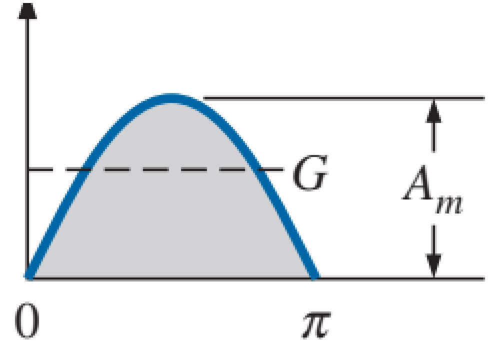
18. Average Value (10/12)

Since we know the area under the positive (or negative) pulse, we can easily determine the average value of the positive (or negative) region of a sine wave pulse by:

$$\text{Average} = G = \frac{2A_m}{\pi}$$

and

$$\boxed{G = \frac{2A_m}{\pi} = 0.637A_m}$$



18. Average Value (11/12)

EXAMPLE 7 Determine the average value of the sinusoidal waveform in Fig. 10.

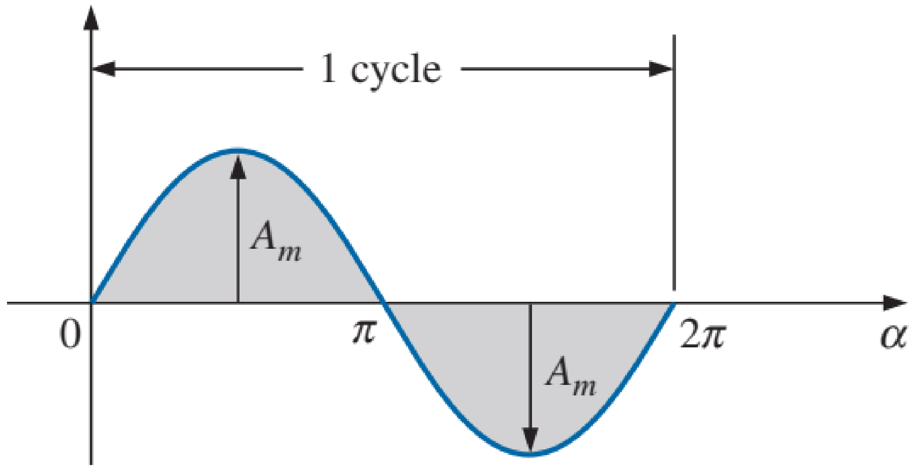


FIG. 10

Solution: By inspection it is fairly obvious that

the average value of a pure sinusoidal waveform over one full cycle is zero.

$$G = \frac{+2A_m - 2A_m}{2\pi} = \mathbf{0\ V}$$

18. Average Value (12/12)

EXAMPLE 8 Determine the average value of the waveform in Fig. 11.

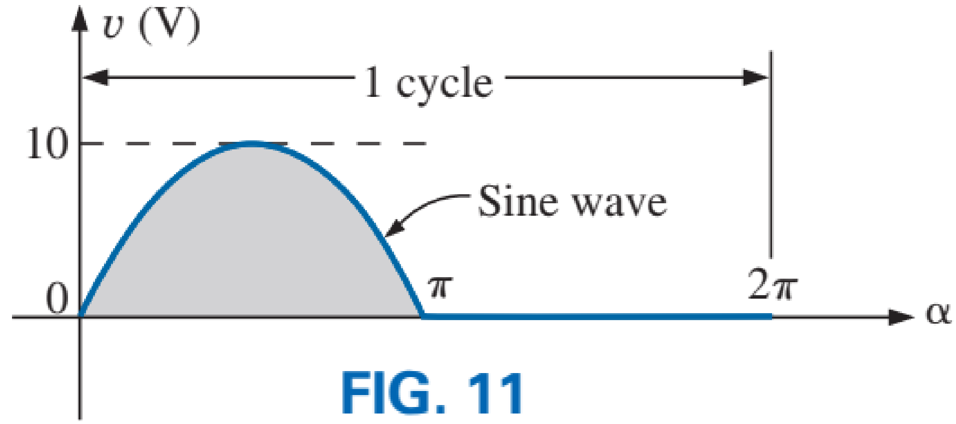


FIG. 11

Solution:

$$G = \frac{2A_m + 0}{2\pi} = \frac{2(10\ \text{V})}{2\pi} \cong \mathbf{3.18\ V}$$

19. Effective (rms) Values (1/9)

- This section begins to relate dc and ac quantities with respect to the power delivered to a load. It will help us determine the amplitude of a sinusoidal ac current required to deliver the same power as a particular dc current.It tells how many volts or amps of dc that an ac waveform supplies in terms of its ability to produce the same average power.
- A fixed relationship between ac and dc voltages and currents can be derived from the experimental setup shown in Fig. 12 next slide. A resistor in a water bath is connected by switches to a dc and an ac supply. If switch 1 is closed, a dc current *I*, determined by the resistance Rand battery voltage *E*, is established through the resistor *R*. The temperature reached by the water is determined by the dc power dissipated in the form of heat by the resistor.

19. Effective (rms) Values (2/9)

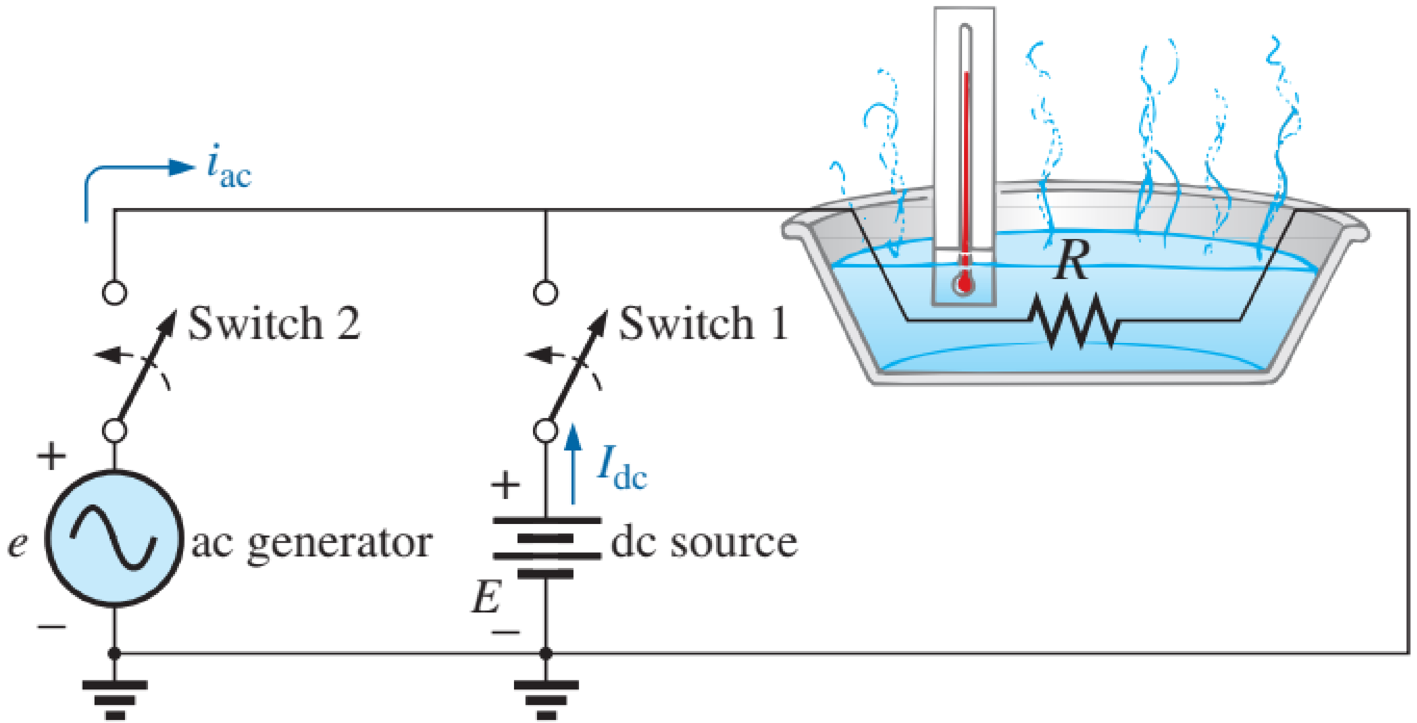


FIG. 12

An experimental setup to establish a relationship between dc and ac quantities.

19. Effective (rms) Values (3/9)

If switch 2 is closed and switch 1 left open, the ac current through the resistor has a peak value of I_m. The temperature reached by the water is now determined by the ac power dissipated in the form of heat by the resistor. The ac input is varied until the temperature is the same as that reached with the dc input. When this is accomplished, the average electrical power delivered to the resistor R by the ac source is the same as that delivered by the dc source.

The power delivered by the ac supply at any instant of time is

$$P_{ac} = (i_{ac})^2 R = (I_m \sin \omega t)^2 R = (I_m^2 \sin^2 \omega t) R$$

However,

$$\sin^2 \omega t = \frac{1}{2}(1 - \cos 2\omega t) \qquad \text{(trigonometric identity)}$$

Therefore,

$$P_{ac} = I_m^2 \left[\frac{1}{2}(1 - \cos 2\omega t) \right] R$$

19. Effective (rms) Values (4/9)

$$\boxed{P_{ac} = \frac{I_m^2 R}{2} - \frac{I_m^2 R}{2}\cos 2\omega t}$$

The *average power* delivered by the ac source is just the first term, since the average value of a cosine wave is zero even though the wave may have twice the frequency of the original input current waveform. Equating the average power delivered by the ac generator to that delivered by the dc source.

$$P_{av(ac)} = P_{dc}$$

$$\frac{I_m^2 R}{2} = I_{dc}^2 R$$

and

$$I_{dc} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

which, in words, states that

the equivalent dc value of a sinusoidal current or voltage is $1/\sqrt{2}$ or 0.707 of its peak value.

19. Effective (rms) Values (5/9)

- The equivalent dc value is called the ***rms (root-mean-square)*** or ***effective value*** of the sinusoidal quantity.
- The relationship between the peak value and the rms value is the same for voltages, resulting in the following set of relationships for the examples and text material to follow:

$$I_{\text{rms}} = \frac{1}{\sqrt{2}} I_m = 0.707 I_m$$

$$E_{\text{rms}} = \frac{1}{\sqrt{2}} E_m = 0.707 E_m$$

- $I_{dc} = I_{rms}$
- $E_{dc} = E_{rms}$

Similarly,

$$I_m = \sqrt{2} I_{\text{rms}} = 1.414 I_{\text{rms}}$$

$$E_m = \sqrt{2} E_{\text{rms}} = 1.414 E_{\text{rms}}$$

19. Effective (rms) Values (6/9)

EXAMPLE 9 Find the rms values of the sinusoidal waveform in each part in Fig. 13

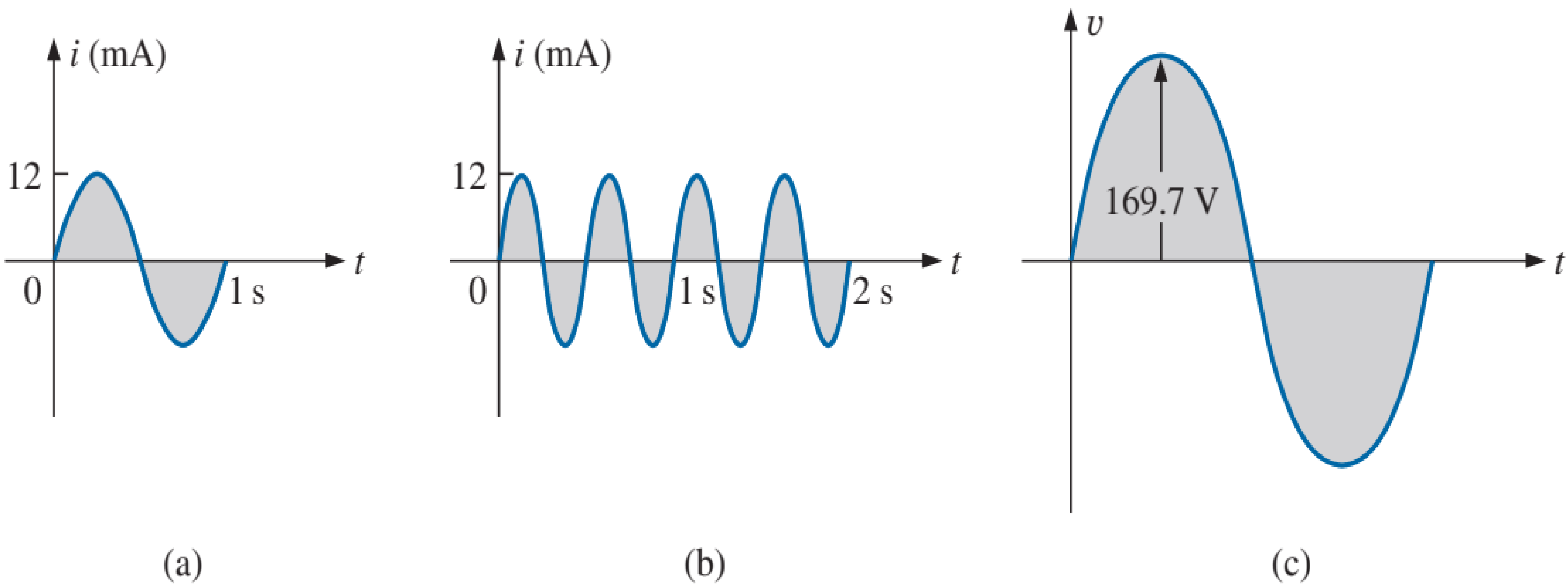


FIG. 13

19. Effective (rms) Values (7/9)

Solution: For part (a), $I_{rms} = 0.707(12 \times 10^{-3}\ A) =$ **8.48 mA.** For part (b), again $I_{rms} =$ **8.48 mA.** Note that frequency did not change the effective value in (b) compared to (a). For part (c), $V_{rms} = 0.707(169.73\ V) \cong$ **120 V,** the same as available from a home outlet.

19. Effective (rms) Values (8/9)

EXAMPLE 10 The 120 V dc source in Fig. 14(a) delivers 3.6 W to the load. Determine the peak value of the applied voltage (E_m) and the current (I_m) if the ac source [Fig. 14(b)] is to deliver the same power to the load.

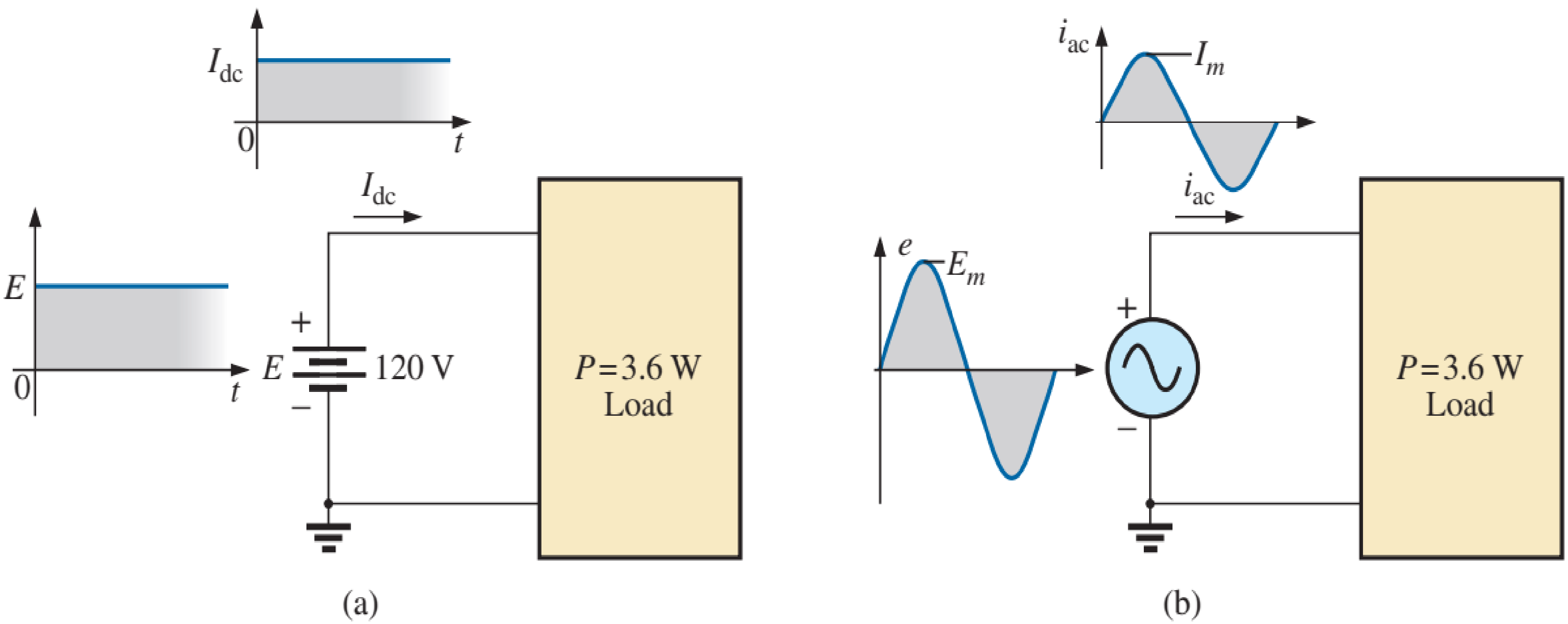


FIG. 14

19. Effective (rms) Values (9/9)

Solution:

$$P_{dc} = V_{dc} I_{dc}$$

and
$$I_{dc} = \frac{P_{dc}}{V_{dc}} = \frac{3.6\ \text{W}}{120\ \text{V}} = 30\ \text{mA}$$

$$I_m = \sqrt{2} I_{dc} = (1.414)(30\ \text{mA}) = \mathbf{42.42\ mA}$$

$$E_m = \sqrt{2} E_{dc} = (1.414)(120\ \text{V}) = \mathbf{169.68\ V}$$

References: Textbooks

1. *Robert L.Boylestad. (2014). Introductory Circuit Analysis. 12th Edition. Pearson Education Limited.*
2. *Allan H.Robbins, Wilhelm C.Miller. (2012). Circuit Analysis Theory and Practice. 5th Edition. Cengage Learning.*
3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6th Edition. McGraw Hill Education.*